A leakage-free validation protocol and mechanism-keyed applicability domain for orientation-resolved ductile damage descriptors in additively manufactured metals

Han Wang[a,*], Lu Zhang[a], Anna Stepashkina[a], Qi Hu[b]

[a] *Zhejiang Lab, 2880 Wenyi West Rd, Hangzhou, 311121, China*
[b] *Department of Plasticity Technology, School of Materials Science and Engineering, Shanghai Jiao Tong University, Shanghai, 200030, China*

Abstract

In additively manufactured (AM) metals, texture and columnar grains make ductile damage orientation-resolved, but descriptor validation is confounded by data leakage and by the iso-strain (Taylor) up-scaling bias, which over-smooths load partitioning, so much predicted anisotropy is a homogenization artefact; accuracy alone cannot tell signal from artefact. We answer with a *leakage-free validation protocol* and an *applicability framework*. *(i) Homogenization bias.* Against spectral-FFT and three-dimensional crystal-plasticity finite-element (CPFEM) full-field references the Taylor surrogate mis-predicts plastic flow by up to $\pm 26\%$, and its fracture-strain bias *sign* is set by the kinematic constraint, not the damage law. *(ii) Direction at indistinguishable accuracy.* Under leakage-free validation the orientation descriptors *cannot be distinguished* from an equal-parameter Lemaitre baseline in blind accuracy — we do not assert that the enriched model beats it — yet they carry a positive but *not statistically significant* ordering signal (Kendall $\bar{\tau} = +0.21$, $p = 0.53$), near-quantitative only in the correctly-polarised columnar sub-domain, not across polarity or in the FCC alloys. *(iii) Applicability boundary.* A per-material, per-mechanism rule states when they transfer, when only directional screening is admissible, and when recalibration is needed. **The contribution is the protocol and the mapped boundary, not a mean-accuracy gain.** The boundary is instantiated as a *characterization-only gate* ($\xi_0 \geq 3$) that routes each alloy to the directional model only where its columnar mechanism is active, otherwise to the isotropic baseline (blind error 15.9% against 24.3% always-isotropic and 31.2% ungated), hence never worse than the baseline by construction.

Keywords: Continuum damage mechanics, Crystal plasticity, Additive manufacturing, Ductile fracture, Multiscale modeling

*Corresponding author.
Email address: wanghan2304@zhejianglab.org (Han Wang)

1. Introduction

1.1. Additive manufacturing of metallic alloys and fracture challenges

Additive manufacturing (AM) of metallic alloys, particularly via laser powder bed fusion (LPBF) and directed energy deposition (DED), has emerged as a transformative manufacturing paradigm for producing geometrically complex components in aerospace, biomedical, and energy industries [1, 2]. However, the widespread adoption of AM metallic components in safety-critical applications is constrained by the limited understanding and predictive capability of their fracture behavior [3].

The layer-by-layer fabrication process inherent to AM produces unique microstructural characteristics that fundamentally differ from conventionally manufactured counterparts [4, 5]:

(i) **Process-inherent defects**: Lack-of-fusion (LOF) pores (10-100 μm), gas-entrapped porosity (5-50 μm), and occasional micro-cracks serve as pre-existing damage nucleation sites.

(ii) **Melt pool boundaries**: Rapid solidification creates chemical segregation and weak interfacial regions at melt pool peripheries, acting as preferential paths for damage propagation.

(iii) **Heterogeneous grain morphology**: The steep thermal gradients produce columnar grains elongated along the build direction (BD), equiaxed grains in remelted zones, and complex duplex structures depending on process parameters.

(iv) **Crystallographic texture**: Epitaxial grain growth during solidification induces pronounced crystallographic texture (e.g., $\langle 001 \rangle$ fiber texture in cubic alloys along BD), leading to mechanical anisotropy.

These microstructural features collectively render the fracture behavior of AM alloys highly anisotropic, process-parameter-dependent, and distinct from wrought/cast alloys [6, 7].

1.2. Current state of fracture modeling for AM alloys

Experimental investigations have extensively documented the anisotropic tensile and fracture properties of AM alloys [8, 9, 10]. However, purely experimental approaches are insufficient for establishing quantitative microstructure-property-fracture relationships necessary for component design and process optimization.

On the numerical modeling front, four families have been pursued, each leaving one of the two requirements of the present problem unmet: a per-family comparison against the model developed here, including the coupling and validation status of each family, is tabulated in Table 5. *Macroscopic phenomenological* criteria — the Johnson-Cook [11] and Bai-Wierzbicki [12] models and related stress-state-dependent ductile fracture criteria [13] — reproduce orientation dependence through fitted per-orientation parameters but carry no microstructural

physics, so their validity ends where their calibration regime does. *Porous plasticity* models of the Gurson-Tvergaard-Needleman (GTN) type [14, 15] evolve a void volume fraction and are physically motivated for ductile void growth and coalescence, yet AM microstructure enters them only as initial porosity and the matrix stays isotropic. *Crystal plasticity (CP)* models resolve grain-level crystallographic slip and reproduce the anisotropic response of AM alloys [16, 17, 18], but most stop at plastic deformation and never evolve damage. *Continuum damage mechanics (CDM)* in the thermodynamic framework [19, 20] gives a systematic account of stiffness degradation and failure and is widely applied in the Lemaitre form [21], but its damage parameters are treated as phenomenological constants with no microstructural sensitivity.

1.3. Motivation: why Taylor homogenization misrepresents orientation-resolved damage

The mechanistic question behind this paper is narrower and more physical than "can another damage law fit AM fracture data": *how is the orientation-resolved competition between crystallographic slip and damage transmission affected when the polycrystal is up-scaled with an iso-strain (Taylor) rule, and does embedding AM microstructural descriptors recover signal that the homogenization filters out?* In these alloys ductile damage is driven by the microscale accumulated plastic strain and stress triaxiality—void growth at process-inherent porosity and melt-pool boundaries, and decohesion along the Si network or prior-β colony interfaces—all of which are set by *which* slip systems a grain activates and *how much* strain localizes in its weakly-oriented neighbours. In a compliant full-field polycrystal the grains deform compatibly but heterogeneously: soft-oriented grains yield early, plastically relax and shed load onto harder neighbours, so the field that drives damage is strongly orientation- and neighbourhood-dependent. The Taylor rule imposes a common strain on every grain and lets only the stress partition; it therefore suppresses the strain heterogeneity that localizes damage and re-distributes the orientation signal through the homogenization rather than resolving it. The bias is not uniform across textures: for a strongly columnar material with one dominant orientation family near the build direction (Ti-6Al-4V) the iso-strain constraint is comparatively benign and a projection of the descriptors can transmit the measured anisotropy almost quantitatively, whereas for weakly textured FCC metals with many comparably-loaded orientation families (316L, AlSi10Mg) the same over-smoothing systematically mis-weights the transverse damage—a structural reason, established independently of any fitted parameter, for the transverse over-projection quantified later in this work. Separating *what the descriptors add* from *what the Taylor up-scaling distorts* is the core of the paper, and it cannot be done from calibration accuracy alone because the orientation signal and the homogenization error are entangled in it.

Three lines of prior work leave this separation unresolved, and each is a symptom of the same mechanistic gap rather than a modelling preference. Existing CDM fracture models for AM either ignore the microstructure or, at best, carry only initial porosity (as a void volume fraction in GTN), so the

combined orientation signature of melt-pool boundaries, grain morphology and texture never enters the driving force *(the descriptor gap)*. Studies that combine crystal plasticity with damage most often decouple the two—running CP first and feeding extracted fields into a macroscopic criterion—which discards the damage-induced degradation of the measured response along the slip path *(the coupling gap)*; in the present work damage enters through a *one-way damage-degraded* coupling that degrades the nominal stress along the CP solution, while the full two-way feedback of a fully coupled formulation is deliberately not claimed (Section 2.5, and the quantified limitation in Section 7). And validation is typically single-material (most often Ti-6Al-4V), which cannot expose the texture-dependence of the homogenization bias described above *(the multi-material gap)*. This work addresses all three within one Taylor-type crystal-plasticity continuum-damage model and reports, under a leakage-free protocol, how much of the fracture anisotropy the descriptors genuinely carry once the Taylor bias is bounded against full-field references.

1.4. Objectives and positioning of the present work

Physically, the objective of this work is to resolve how orientation-dependent crystallographic slip and ductile damage interact once an additively manufactured columnar/equiaxed polycrystal is up-scaled with an iso-strain (Taylor) rule, and whether embedding AM microstructural descriptors in the damage law recovers orientation signal that the homogenization filters out. Methodologically, the contribution is the validation protocol itself: we construct a leakage-free framework—identical fitted-parameter count, training data and cross-validation for the descriptor-based surrogate and a classical Lemaitre baseline—that isolates the descriptor signal from the homogenization artefact. Applied to nine calibration cases and four independent blind datasets, the framework shows that the descriptors are statistically indistinguishable from the baseline in blind accuracy yet carry direction information the isotropic baseline lacks, and it converts prediction residuals that would otherwise remain unexplained into a mapped, mechanism-specific applicability boundary that a follow-on correction can target. Its purpose is to establish a *reusable, leakage-free protocol* and to characterize *when* the descriptors help; the paired comparison is inherently underpowered at the handful of available public as-built datasets, and its minimum detectable effect and strict *failure-to-distinguish* (never demonstrated-equivalence) reading are quantified in Section 5.5.

This work makes the following contributions:

(1) **A leakage-free validation protocol that separates descriptor signal from homogenization artefact (the methodological contribution)**: The modelling contributions below are tested with a four-tier verification framework (code convergence, full-field homogenization-bias quantification, strict calibration/validation separation, and independent literature validation) and a leakage-free leave-one-orientation-out cross-validation protocol in which orientation multipliers are removed entirely and the single damage strength is re-calibrated per fold (Section 5.5).

Four independent blind sets from separate publications (Inconel 718, and second datasets for Ti-6Al-4V, 316L and AlSi10Mg, Sections 7.4–7.5) are systematically classified by failure mechanism into an applicability domain with a decision procedure (Section 7.2). The reusable leakage-free validation methodology and the mechanism-keyed applicability boundary it yields are the central contribution; that boundary is instantiated as a characterization-only, leakage-free gate ($\xi_0 \geq 3$; Section 5.6, Table 3) that routes each alloy to the directional model or to the isotropic baseline, adds no fitted parameter, and is therefore never worse than the baseline by construction.

(2) **Orientation-resolved slip–damage coupling and its Taylor-homogenization sensitivity (the plasticity contribution)**: A Taylor-type crystal-plasticity continuum-damage model is built in which a damage evolution equation explicitly incorporates five physically-motivated AM microstructural descriptors—porosity volume fraction (ϕ), initial damage (D_0), melt pool boundary density (λ), grain morphology factor (ξ), and texture orientation weight (θ)—through the multiplicative AM correction f_{AM}, applied as a one-way damage-degraded coupling along the slip path. The contribution is mechanistic: it separates *what these orientation-resolved descriptors add* from *what the iso-strain up-scaling distorts*, showing that the descriptors carry directional information (they match the strongly columnar Ti-6Al-4V direction ratio and the measured direction ordering, not high absolute accuracy) while the Taylor bias that filters the orientation signal is bounded against full-field spectral-FFT and CPFEM references (Section 7.3).

(3) **One-way damage-degraded CP-CDM coupling with quantified homogenization bias**: The modified damage model is implemented at the constitutive level in the in-house TaylorCPCDM solver, which shares DAMASK's constitutive interface conventions; DAMASK's spectral-FFT solver serves exclusively as a full-field reference quantifying the Taylor iso-strain damage-free UTS bias (Section 7.3; per-alloy values reported there and in Section 5). The formulation is one-way coupled and damage-degraded—damage accumulates along the CP-computed path and degrades the nominal (measured) stress while the slip-driving stress is evaluated on the undamaged stiffness (Section 2.5); a full two-way feedback is not claimed. The coupled-damage fracture-strain bias is quantified against in-house two- and three-dimensional full-field CPFEM references, whose *sign* is set by the kinematic constraint and whose *magnitude* by the failure criterion rather than a single converged number (Section 7.3; full breakdown in Supplementary Section S1).

(4) **Systematic multi-material calibration with an explicit generalization assessment**: The model is calibrated against literature-based experimental data for three AM alloys spanning distinct crystal structures: HCP Ti-6Al-4V (columnar grains), FCC 316L stainless steel (cellular/equiaxed), and FCC AlSi10Mg (Si network with duplex morphology). Each material is tested at three build orientations (0°, 45°, 90°), yield-

ing nine calibration cases; generalization is assessed separately with the leakage-free cross-validation protocol (Section 5.5).

(5) **An end-to-end standardized simulation workflow**: We establish an automated pipeline from literature-based microstructural descriptors to fracture prediction, integrating RVE generation, deterministic single-parameter calibration, and automated post-processing. The workflow is solver-agnostic and demonstrated with the TaylorCPCDM framework, with the DAMASK full-field solver employed for the homogenization-bias assessment.

Contribution in fracture-mechanics terms.. Read as a contribution to fracture mechanics rather than to materials modelling, the deliverable is a ductile-damage descriptor *together with the conditions under which it may be used quantitatively*. Three statements belong to that vocabulary. First, damage is driven by the thermodynamic force $Y = \tilde{\sigma}^2/2E$ of the damaged aggregate evaluated along the crystal-plasticity solution, so the orientation dependence of damage nucleation at process-inherent porosity and melt-pool boundaries enters the evolution law through measured microstructural descriptors (ϕ, λ, ξ, θ) rather than through fitted per-orientation coefficients — a mechanism-level rather than a curve-fitting account of ductility anisotropy. Second, the anisotropy the model reproduces is the $0°/90°$ *fracture-strain* contrast and the direction of the damage rate, i.e. the ductility ordering that governs where and when a crack initiates in an as-built component; the formulation does not produce a fracture toughness or a J-resistance curve, and no such value is reported or implied here. Third, the applicability boundary is expressed in quantities a metallography or fracture laboratory measures routinely — grain aspect ratio and melt-pool-boundary line density — so it can be evaluated *before* a specimen is machined, and the quantified homogenization bias bounds how far the resulting low-order directional statement may be trusted when it is used to rank build orientations for flaw assessment.

The remainder of this paper is organized as follows. Section 2 presents the theoretical framework, including the CDM foundation, the proposed AM-specific modifications, and the CP constitutive model. Section 3 describes the multiscale coupling framework and DAMASK implementation. Section 4 details the standardized simulation workflow. Section 5 presents the systematic calibration against literature data, the independent leave-one-orientation-out cross-validation, and the applicability-gated hybrid that routes each alloy by its columnar aspect ratio. Section 7 discusses the model applicability, limitations, and comparison with alternative approaches. Section 8 summarizes the key findings and engineering implications.

2. Model Development

2.1. Thermodynamic framework of continuum damage mechanics

The CDM framework is constructed within the thermodynamics of irreversible processes with internal variables [19]. The state of a material point

is described by:

- Elastic strain tensor $\boldsymbol{\varepsilon}^e$ (observable variable)
- Accumulated plastic strain p (internal variable)
- Isotropic damage variable D (internal variable, $0 \leq D \leq 1$)

State potential and damage-energy coupling

The Helmholtz free energy, incorporating elastic-damage coupling, is postulated as:

$$\rho\psi(\boldsymbol{\varepsilon}^e, D) = \frac{1}{2}(1-D)\,\boldsymbol{\varepsilon}^e : \mathbb{C} : \boldsymbol{\varepsilon}^e \tag{1}$$

where $\mathbb{C}$ is the fourth-order elastic stiffness tensor of the undamaged material. The thermodynamic force conjugate to damage—the damage energy release rate—follows from the state potential:

$$Y = -\rho\frac{\partial\psi}{\partial D} = \frac{1}{2}\boldsymbol{\varepsilon}^e : \mathbb{C} : \boldsymbol{\varepsilon}^e \tag{2}$$

The effective stress concept, rooted in the hypothesis of strain equivalence, relates the nominal Cauchy stress to an effective stress acting on the undamaged load-bearing area:

$$\tilde{\boldsymbol{\sigma}} = \frac{\boldsymbol{\sigma}}{1-D} \tag{3}$$

In the present implementation the effective (undamaged-stiffness) stress is used for the damage energy release rate of Eq. (2); the flow rule operates on the same undamaged-stiffness stress (Section 2.5), so damage exerts no direct rescaling on plastic flow, and the nominal $(1-D)$ factor of Eq. (26) is confined to the measured macroscopic stress.

Dissipation and evolution laws

The intrinsic mechanical dissipation inequality reads:

$$\mathcal{D} = \boldsymbol{\sigma} : \dot{\boldsymbol{\varepsilon}}^p - R\dot{p} + Y\dot{D} \geq 0 \tag{4}$$

where R is the isotropic hardening thermodynamic force. Following the generalized normality rule, the evolution of damage is driven by the damage energy release rate Y above a threshold:

$$\dot{D} = \frac{\partial\Phi^*}{\partial Y}\cdot\dot{p} \tag{5}$$

where Φ^* is the dissipation potential.

2.2. Conventional Lemaitre-type damage model

In the classical Lemaitre formulation [21, 22], the dissipation potential for damage is chosen as:

$$\Phi_D^* = \frac{S}{(s+1)} \left(\frac{Y}{S} \right)^{s+1} \tag{6}$$

yielding the widely-adopted damage evolution equation:

$$\dot{D} = \left(\frac{Y}{S} \right)^{s} \cdot \dot{p} \cdot H(p - p_D) \tag{7}$$

where S is the damage energy strength parameter, s is the damage exponent, p_D is the damage threshold plastic strain, and $H(\cdot)$ is the Heaviside step function. While effective for conventional wrought/cast alloys, Eq. (7) possesses three critical limitations when applied to AM alloys:

(i) The parameters S and s are treated as material constants, independent of microstructural variability.
(ii) There is no provision for initial (pre-existing) damage.
(iii) The isotropic formulation cannot capture orientation-dependent damage evolution arising from crystallographic texture and grain morphology anisotropy.

2.3. Proposed AM-specific modification of the damage evolution equation

Physical basis for modification

The proposed modification is grounded in the five physically-identifiable AM microstructural descriptors—porosity ϕ, initial damage D_0, melt-pool-boundary density λ, grain-morphology factor ξ and texture orientation weight θ—presented below as four feature groups (porosity and initial damage share one item), each with a well-established influence on damage mechanisms:

(a) Initial porosity and pre-existing damage D_0: AM processes inevitably introduce volumetric defects. XCT characterization reveals total porosity fractions typically in the range 0.1–2 % for optimized LPBF parameters, and significantly higher for sub-optimal conditions [23]. These defects serve as pre-existing damage nucleation sites, motivating the introduction of a non-zero initial damage variable:

$$D_0 = f(\rho_{\mathrm{LOF}}, \rho_{\mathrm{gas}}, \mathrm{VED}) \tag{8}$$

where ρ_{LOF} and ρ_{gas} are the volume fractions of lack-of-fusion and gas porosity, respectively, and VED is the volumetric energy density linking process parameters to defect population.

(b) Melt pool boundary density λ: Melt pool boundaries (MPBs) are regions of elemental micro-segregation, dislocation accumulation, and occasionally micro-void coalescence [4]. They constitute mechanically weaker interfaces that facilitate damage propagation. We define the MPB density as:

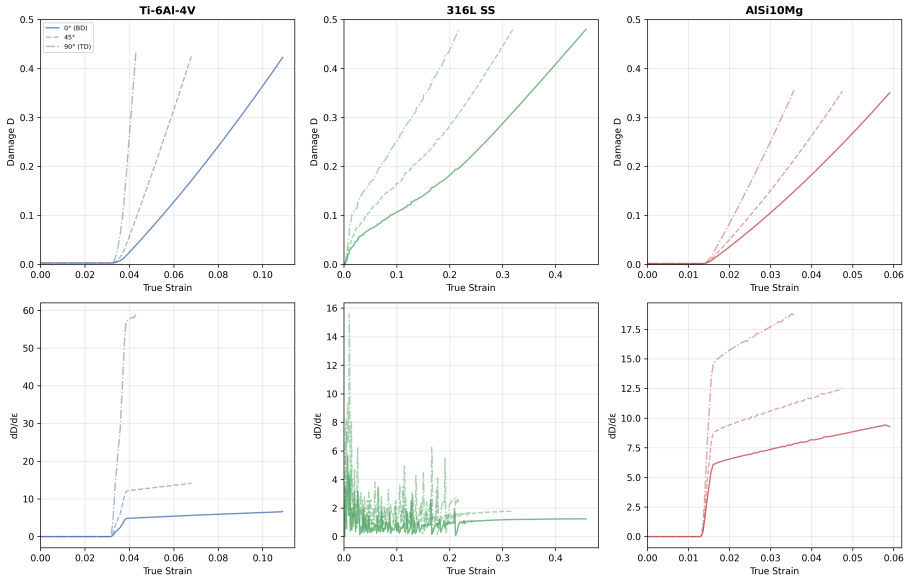


Figure 1: Orientation-dependent damage response that the AM descriptors are designed to capture: predicted damage D (top row) and damage rate $\mathrm{d}D/\mathrm{d}\varepsilon$ (bottom row) versus true strain for the three AM alloys (Ti-6Al-4V, 316L SS, AlSi10Mg) at loading angles 0° (build direction), 45° and 90° (transverse direction). For every alloy the transverse 90° orientation accumulates damage fastest, reaching a given damage level at the smallest strain—the orientation-dependent acceleration that the descriptors introduced below are constructed to represent.

$$\lambda = \frac{\text{Total MPB interfacial area}}{\text{Volume}} \tag{9}$$

Quantifiable from optical/SEM metallography through intercept counting, λ typically ranges from 10–100 /mm depending on hatch spacing and layer thickness.

(c) Grain morphology factor ξ: The aspect ratio of columnar grains directly influences the available slip length and stress concentration at grain boundaries [6]. We define:

$$\xi = \frac{L_{\max}}{L_{\min}} \tag{10}$$

where $L_{\max}$ and $L_{\min}$ are the major and minor axes of the grain, respectively. For equiaxed grains $\xi \approx 1$; for highly columnar grains in AM alloys, ξ can reach 5–10.

(d) Texture orientation weight θ: The crystallographic texture determines the ease of slip system activation, which in turn governs plastic strain accommodation and damage nucleation. We quantify the texture by the *texture weight* θ, defined as the fraction of grains belonging to the $\langle 001 \rangle$-type fiber

texture:

$$\theta = \frac{N_{\langle 001 \rangle}}{N_g} \tag{11}$$

where $N_{\langle 001 \rangle}$ is the number of grains whose $\langle 001 \rangle$ axis lies within a tolerance of the build direction, and N_g is the total number of grains. θ is directly measurable by EBSD pole-figure analysis (Table 1). The orientation effect on damage enters f_{AM} through the *orientation-resolved mean Schmid factor* $\bar{\theta}(\psi)$ in Eq. (15): for a polycrystal sampled with texture weight θ (Section 3), $\bar{\theta}(\psi)$ is the ensemble average of the maximum absolute Schmid factor resolved along the loading direction ψ. Higher $\bar{\theta}(\psi)$ values correspond to softer orientations with easier slip activation, typically associated with delayed damage initiation.

Modified damage evolution equation

The proposed modified damage evolution equation takes the form:

$$\dot{D} = \left(\frac{Y}{S_0}\right)^s \cdot \dot{p} \cdot f_{\mathrm{AM}}(\phi, \theta, D_0, \lambda, \xi; \psi) \cdot H(p - p_D) \tag{12}$$

with the initial condition $D(t = 0) = D_0$. Since the free energy Eq. (1) is written per unit volume ($\rho\psi$), the thermodynamic force Y of Eq. (2) is an elastic energy density with the dimensions of stress (Pa, quoted here in MPa); the damage energy strength S_0 therefore carries the same units, so the ratio Y/S_0 is dimensionless as the exponent s requires. For the calibrated Ti-6Al-4V value $S_0 = 0.32\,\mathrm{MPa}$ the peak energy release rate $Y \sim \sigma_u^2/2E$ is of order a few MPa over the loading range, keeping Y/S_0 of order unity to ten—the adopted unit and value are dimensionally consistent.

Remark on parameter bookkeeping. To avoid double counting of the same microstructural descriptors, the damage energy strength is *not* modified by λ or ξ in the present formulation: the baseline parameter S_0 is a material constant calibrated exclusively on the 0° orientation data, and *all* microstructural and orientation effects enter the model solely through the dimensionless AM correction function f_{AM}. Early versions of the model that simultaneously scaled $S(\lambda, \xi) = S_0 \cdot g_1(\lambda) \cdot g_2(\xi)$ and multiplied f_{AM} by $f_\lambda \cdot f_\xi$ double-counted the melt-pool-boundary and grain-morphology descriptors and produced directionally inconsistent effects (increasing S retards damage while $f_\lambda, f_\xi > 1$ accelerate it). That redundancy has been removed; $S \equiv S_0$ throughout.

The **AM correction function** f_{AM} multiplicatively combines the five microstructural effects (with a sixth geometric-shielding factor added for the defect-shielding blind alloys):

$$f_{\mathrm{AM}}(\phi, \theta, D_0, \lambda, \xi; \psi) = f_\phi \cdot f_\theta \cdot f_{D_0} \cdot f_\lambda(\psi) \cdot f_\xi(\psi) \cdot f_{\mathrm{shield}}(\psi) \tag{13}$$

with each sub-function defined as:

$$f_\phi = 1 + \beta_1 \left(\frac{\phi}{\phi_{\text{crit}}} \right)^{m_1} \tag{14}$$

$$f_\theta = 1 + \beta_2 \left[\max\!\left(1 - 2\bar{\theta}(\psi),\, 0\right) \right]^{m_2} \tag{15}$$

$$f_{D_0} = \exp(\beta_3 D_0) \tag{16}$$

$$f_\lambda(\psi) = 1 + \beta_4 \left(\bar{\lambda}(\psi) - 1 \right) \tag{17}$$

$$f_\xi(\psi) = 1 + \beta_5 \left(1/\bar{\xi}(\psi) - 1 \right) \tag{18}$$

$$f_{\text{shield}}(\psi) = 1 - \beta_6 \sin^2 \psi \tag{19}$$

where the six factors model, respectively, porosity stress concentration, hard-orientation acceleration, initial-damage acceleration, melt-pool-boundary density, transverse short-ligament geometry, and geometric shielding of layered defect facets; each β_i is dimensionless. Because $\bar{\theta}(\psi)$ is the ensemble mean of the maximum Schmid factor resolved on a slip system, it is bounded by the Schmid limit $0 \le \bar{\theta} \le 0.5$, so $1 - 2\bar{\theta}(\psi) \ge 0$ and the non-integer power m_2 of Eq. (15) is always taken of a non-negative base; the explicit $\max(\cdot, 0)$ is a numerical guard against round-off pushing $\bar{\theta}$ marginally above 0.5 (the reference texture-free state $\bar{\theta} = 0.5$ then gives $f_\theta = 1$, so the guard leaves the reference limit unchanged).

Here ϕ is the porosity volume fraction, ϕ_{crit} a critical porosity threshold, and $\beta_{1-6}, m_{1,2}$ are dimensionless calibration coefficients. All descriptors entering f_{AM} are normalized to dimensionless form: $\bar{\lambda}(\psi) = \lambda_{\text{eff}}(\psi)/\lambda_{\text{ref}}$ with reference MPB density $\lambda_{\text{ref}} = 50\,/\text{mm}$, and $\bar{\xi}(\psi) = \xi_{\text{eff}}(\psi)$ is the orientation-resolved aspect ratio of an equivalent spheroidal grain projected along the loading axis (Section 3). The reference state is defined as fully dense, equiaxed, texture-free material ($\phi \to 0$, $\lambda_{\text{eff}} = \lambda_{\text{ref}}$, $\bar{\xi} = 1$, $\bar{\theta} = 0.5$); by construction every sub-function then evaluates to unity—in particular $f_\lambda = 1$ because $\bar{\lambda} = 1$—so $f_{\text{AM}} \to 1$ and Eq. (12) recovers the classical Lemaitre form. The orientation dependence is carried entirely by the physically-motivated projection laws $\lambda_{\text{eff}}(\psi)$, $\xi_{\text{eff}}(\psi)$ and the mean Schmid factor $\bar{\theta}(\psi)$; no empirical orientation multiplier enters the production model. The geometric-shielding factor $f_{\text{shield}}(\psi)$ (Eq. (19)) targets a competing mechanism: layered defect facets (lack-of-fusion boundaries, crack-prone interface networks) whose normals cluster parallel to the build direction, so the facets themselves lie in planes perpendicular to it. Loading parallel to the build direction ($\psi = 0°$), i.e. normal to these facets, opens them directly in mode I, while loading perpendicular to the build direction ($\psi = 90°$), i.e. within the facet planes, shears rather than opens them and is geometrically shielded; f_{shield} therefore counteracts the transverse ($\psi = 90°$) damage acceleration that the projection laws would otherwise impose, which in the defect-shielding blind alloys makes $\psi = 90°$ the more ductile direction. For the calibration alloys, whose fracture is columnar-morphology-dominated, $\beta_6 = 0$ (so $f_{\text{shield}} \equiv 1$ and the production numbers of Section 5 are untouched); $\beta_6 > 0$ is applied only to the defect-shielding blind alloys of Section 7.5, as an assumed, per-material coefficient that is never fitted (hence it adds no degrees of freedom; provenance

class A of Table 1). *We flag at the point of definition that β_6 enters this work only as a post-hoc diagnostic for the single genuine polarity-reversal case (the Hitzler 316L set, Section 7.5): no β_6 value enters any blind-prediction statistic, the LOOCV averages, or the model-selection comparison, and its role is confined to demonstrating the existence of a defect-shielding mechanism, not to improving headline accuracy.*

The exponential form of f_{D_0} reflects the compounding effect of initial damage: pre-existing voids accelerate subsequent damage accumulation by concentrating stress in their vicinity.

Orientation-resolved projection laws

The orientation dependence of the descriptors entering $f_\lambda(\psi)$, $f_\xi(\psi)$ and f_θ is obtained from closed-form geometric projections that carry the full physics of the AM microstructure (no fitted direction parameters; Section 3).

Effective melt-pool-boundary density. The melt pool boundaries and intercolumnar interfaces form a transversely isotropic weak-interface network whose normals lie predominantly in the plane perpendicular to the build direction. Loading at angle ψ to the build direction, the weak-interface area density resolved perpendicular to the loading axis is

$$\lambda_{\text{eff}}(\psi) = \lambda_0 \left[w_{\text{iso}} + (1 - w_{\text{iso}}) \sin^2 \psi \right], \tag{20}$$

where λ_0 is the nominal MPB density (Table 1) and $w_{\text{iso}} = 0.25$ is the isotropic area fraction of the interface network (stereological assumption). At $\psi = 0$ only the isotropic fraction is directly loaded; at $\psi = 90°$ the entire network faces a normal tensile component.

Effective grain-aspect-ratio. Each grain is modelled as a spheroid of revolution about the build direction with semi-axes $(a, a, c = \xi_0 a)$, where ξ_0 is the nominal aspect ratio of Table 1. In the plane containing the build direction and the loading axis, the grain extent along the loading direction is the radial distance from the grain centre to its surface along the loading line, $L(\psi) = 2a\xi_0 / \sqrt{\xi_0^2 \sin^2 \psi + \cos^2 \psi}$, and the extent perpendicular to it is the conjugate radius $W(\psi) = 2a\xi_0 / \sqrt{\xi_0^2 \cos^2 \psi + \sin^2 \psi}$, so that the orientation-resolved aspect ratio is

$$\xi_{\text{eff}}(\psi) = \frac{L(\psi)}{W(\psi)} = \sqrt{\frac{\xi_0^2 \cos^2 \psi + \sin^2 \psi}{\xi_0^2 \sin^2 \psi + \cos^2 \psi}}, \tag{21}$$

with $\xi_{\text{eff}}(0) = \xi_0$, $\xi_{\text{eff}}(45°) = 1$ (the two conjugate radii coincide, so the grain projection is aspect-neutral at 45°) and $\xi_{\text{eff}}(90°) = 1/\xi_0$ (transverse short-axis dominated morphology).

Orientation-resolved mean Schmid factor. To avoid overloading the symbol, we reserve θ for the dimensionless texture-weight *descriptor*—a sampling probability in $[0, 1]$, the fifth entry of Table 1—and $\bar{\theta}(\psi)$ for the *resolved mean Schmid factor* that this texture induces; the correction f_θ is named after the orientation/texture effect but is a function of $\bar{\theta}(\psi)$, not of θ itself. The texture weight θ of Table 1 is sampled in the aggregate generation (Section 3):

with probability θ a grain orientation is drawn from the $\langle 001 \rangle$-type fiber texture about the build direction, otherwise from the Haar-uniform measure on SO(3). The mean Schmid factor entering f_θ is then computed directly from the sampled orientation ensemble $\{\mathbf{R}_g\}_{g=1}^{N_{\mathrm{gr}}}$ as

$$\bar{\theta}(\psi) = \frac{1}{N_{\mathrm{gr}}} \sum_{g=1}^{N_{\mathrm{gr}}} \max_{\alpha} \left| (\mathbf{R}_g \hat{\mathbf{e}}_\psi) \cdot [\mathbf{s}^\alpha \otimes \mathbf{m}^\alpha] \cdot (\mathbf{R}_g \hat{\mathbf{e}}_\psi) \right| \tag{22}$$

where $\hat{\mathbf{e}}_\psi = (\sin\psi, 0, \cos\psi)$ is the loading direction in the sample frame and the maximum runs over all slip systems α. The projection laws of Eqs. (20)–(22) carry the entire direction dependence of the production protocol; no orientation multiplier is fitted.

Physical interpretation of correction functions

Each component of f_{AM} has a clear physical interpretation:

- f_ϕ: Captures stress concentration around pores. As porosity approaches the critical threshold, damage evolution accelerates rapidly. For fully-dense (wrought) material, $\phi \approx 0$, $f_\phi \approx 1$, recovering the classical formulation.
- f_θ: Encodes the crystallographic orientation effect. Materials loaded along soft orientations (high θ) experience more homogeneous plastic deformation, delaying damage localization. Loading along hard orientations (low θ) concentrates strain in fewer favorably-oriented grains, accelerating damage.
- f_{D_0}: Reflects the compounding nature of damage. Pre-existing voids not only reduce the initial load-bearing area but also serve as stress concentrators that promote further void nucleation and growth.
- f_λ: Accounts for the increased probability of damage propagation along weakened melt pool boundaries as their density increases.
- f_ξ: Represents the anisotropic damage susceptibility arising from elongated grain morphology, where elongated grains in the loading direction provide longer uninterrupted slip paths.

Directional slip-resistance correction $g_s(\psi)$

The projection laws of Eqs. (20)–(22) carry the direction dependence of the *damage* evolution. To address the residual gap they leave for the direction dependence of *strength* (the predicted $0°/90°$ fracture-strain ratio compresses the measured ratio, Section 5.3), we introduce a direction-dependent modification of the Voce saturation resistance that preserves the single-parameter protocol. The directional saturation resistance entering the aggregate is

$$\hat{g}_s(\psi) = g_s \left[1 - k_g \left(1 - \frac{\xi_{\mathrm{eff}}(\psi)}{\xi_0} \right) \right], \tag{23}$$

where $\xi_{\text{eff}}(\psi)/\xi_0 = \frac{1}{\xi_0}\sqrt{\frac{\xi_0^2\cos^2\psi+\sin^2\psi}{\xi_0^2\sin^2\psi+\cos^2\psi}}$ follows directly from the projection of Eq. (21), so that $\hat{g}_s(0°) = g_s$, $\hat{g}_s(45°) = g_s\,[1-k_g(1-\xi_0^{-1})]$ (since $\xi_{\text{eff}}(45°) = 1$) and $\hat{g}_s(90°) = g_s\,[1-k_g(1-\xi_0^{-2})]$: transverse loading softens the aggregate when the columnar grains are elongated along the build direction. The coefficient k_g is an assumed, shared parameter ($k_g = 0.15$, provenance class A of Table 1); it is not fitted to fracture data, so the fitted-parameter count remains a single S_0 per material. Section 5.8 verifies that k_g is a low-sensitivity parameter (span 0.4% at 90°, identically zero at 0° because $\hat{g}_s(0°) = g_s$), and Section 5.3 reports the resulting calibration change.

2.4. Crystal plasticity constitutive model

Kinematics

The deformation gradient is multiplicatively decomposed into elastic and plastic parts:

$$\mathbf{F} = \mathbf{F}^e \cdot \mathbf{F}^p \tag{24}$$

where $\mathbf{F}^e$ represents elastic stretching and rigid-body rotation, and $\mathbf{F}^p$ captures the incompressible plastic deformation due to crystallographic slip. The plastic velocity gradient in the intermediate (relaxed) configuration is:

$$\mathbf{L}^p = \dot{\mathbf{F}}^p \cdot (\mathbf{F}^p)^{-1} = \sum_{\alpha=1}^{N_s} \dot{\gamma}^\alpha (\mathbf{s}^\alpha \otimes \mathbf{m}^\alpha) \tag{25}$$

where $\dot{\gamma}^\alpha$ is the shear rate on slip system α, $\mathbf{s}^\alpha$ and $\mathbf{m}^\alpha$ are the slip direction and slip plane normal (unit vectors) in the intermediate configuration, and N_s is the total number of slip systems ($N_s = 12$ for FCC, up to 30 for HCP).

Elastic constitutive law

The second Piola-Kirchhoff stress in the intermediate configuration, incorporating damage-induced stiffness degradation, is:

$$\mathbf{S} = (1-D)\,\mathbb{C} : \mathbf{E}^e \tag{26}$$

where $\mathbf{E}^e = \frac{1}{2}(\mathbf{F}^{eT}\mathbf{F}^e - \mathbf{I})$ is the Green-Lagrange elastic strain. Under the strain-equivalence hypothesis [21], the elastic predictor of the single-crystal update is evaluated with the *undamaged* stiffness, so that the stress driving both plastic flow and damage is computed as if acting on the undamaged load-bearing area:

$$\tilde{\mathbf{S}} = \mathbb{C} : \mathbf{E}^e \tag{27}$$

The resolved shear stress on slip system α is evaluated from this undamaged-stiffness stress:

$$\tilde{\tau}^\alpha = \tilde{\mathbf{S}} : (\mathbf{s}^\alpha \otimes \mathbf{m}^\alpha) \tag{28}$$

By construction $\tilde{\tau}^\alpha$ does *not* depend on the current damage D: damage exerts no direct influence on plastic flow, consistent with the alternative branch of the strain-equivalence framework in which the flow rule operates on the effective (undamaged-stiffness) stress rather than on a damage-rescaled one. The damage-degraded nominal stress of Eq. (26) is used *only* for the measured macroscopic output, through the $(1-D)$ factor applied to the volume-averaged Cauchy stress, which produces the observed post-peak softening. This choice deliberately avoids the internally inconsistent combination in which scaling both $\tilde{\tau}^\alpha$ and g^α by $1/(1-D)$ would leave the slip-driving ratio unchanged; here the damage influence on the macroscopic response enters solely through (i) the nominal $(1-D)$ softening of the measured stress and (ii) the damage evolution and fracture criterion of Section 2.

Flow rule

The shear rate on each slip system follows a power-law rate-dependent formulation in terms of the undamaged-stiffness resolved shear stress:

$$\dot{\gamma}^\alpha = \dot{\gamma}_0 \left| \frac{\tilde{\tau}^\alpha}{g^\alpha} \right|^n \operatorname{sign}(\tilde{\tau}^\alpha) \tag{29}$$

where $\dot{\gamma}_0$ is the reference shear rate, g^α is the slip resistance of system α (which is *not* scaled by damage), and n is the rate sensitivity exponent. The limit $n \to \infty$ corresponds to rate-independent behavior. The driving ratio $\tilde{\tau}^\alpha / g^\alpha$ is independent of D, so damage does not promote or retard yielding at the constitutive level; the post-peak softening observed in the measured macroscopic response originates entirely from the $(1-D)$ factor of Eq. (26) applied to the volume-averaged Cauchy stress.

Hardening evolution

The evolution of slip resistance follows a Voce-type hardening law:

$$\dot{g}^\alpha = \sum_{\beta=1}^{N_s} h_{\alpha\beta} |\dot{\gamma}^\beta| \tag{30}$$

with the hardening matrix:

$$h_{\alpha\beta} = q_{\alpha\beta} \cdot h_0 \cdot \left| 1 - \frac{g^\beta}{g_s} \right|^a \cdot \operatorname{sign}\left(1 - \frac{g^\beta}{g_s} \right) \tag{31}$$

where h_0 is the initial hardening modulus, g_s is the saturation slip resistance, a is the hardening exponent, and $q_{\alpha\beta}$ is the latent hardening matrix ($q_{\alpha\beta} = 1$ for coplanar systems $\alpha = \beta$, and $q_{\alpha\beta} = q_{\text{lat}}$ for non-coplanar systems). The initial condition is $g^\alpha(0) = g_0$ (initial CRSS).

Damage does *not* rescale the slip resistance: the hardening law (Eq. (30)) evolves g^α independently of D, and the resolved shear stress of Eq. (28) is likewise evaluated on the undamaged stiffness, so damage has *no direct influence on plastic flow*. The influence of damage on the macroscopic response enters solely

through (i) the nominal $(1-D)$ factor of Eq. (26) applied to the measured stress and (ii) the damage evolution and fracture criterion of Section 2. This is the second branch of the coupling ambiguity identified above: rather than rescaling both the resolved shear stress and the slip resistance by $1/(1-D)$ (which would cancel the damage effect entirely), we explicitly state that the flow rule operates on the undamaged-stiffness stress and that damage affects the macroscopic response only through the nominal softening and the damage/fracture laws.

2.5. Coupling between crystal plasticity and damage

The CP-CDM coupling is implemented as a **one-way damage-degraded CP-CDM** scheme (plasticity drives damage; damage degrades the load-bearing stiffness but does not feed back on the flow rule). At each time increment $[t, t+\Delta t]$, the coupled system is solved:

(1) **Elastic predictor**: Assume no plastic flow, compute the trial stress based on $\mathbf{F}(t+\Delta t)$ and the current damage $D(t)$.
(2) **Yield check**: Evaluate the yield criterion. If $f \leq 0$, the step is elastic; proceed to step 5.
(3) **Plastic corrector (Newton-Raphson)**: If $f > 0$, solve the nonlinear system:

$$\Delta\gamma^{\alpha} = \dot{\gamma}^{\alpha}(\tau^{\alpha}, g^{\alpha}) \cdot \Delta t \tag{32}$$

$$g^{\alpha} \leftarrow g^{\alpha} + \Delta g^{\alpha} \tag{33}$$

$$D \leftarrow D + \Delta D \tag{34}$$

where the second line updates the slip-system hardening and the third the damage via Eq. (12).
(4) **Stress update**: Recompute the nominal (damage-degraded) Cauchy stress with updated damage:

$$\boldsymbol{\sigma} = (1-D)\,\mathbb{C} : (\mathbf{E}^{e} - \Delta\mathbf{E}^{p}) \tag{35}$$

which is used exclusively for the measured macroscopic output; the plastic-flow and damage drivers (Eqs. (28), (12)) use the undamaged-stiffness stress $\tilde{\boldsymbol{\sigma}} = \mathbb{C} : \mathbf{E}^{e}$.
(5) **State output**: $\mathbf{F}^{p}$, g^{α}, D, $\boldsymbol{\sigma}$.

The damage variable affects the CP computation through three mechanisms, each expressed at the constitutive level:

- **Nominal stiffness degradation (output only)**: $\mathbb{C}^{\text{eff}} = (1-D)\mathbb{C}$ enters the nominal stress of Eq. (26), which is used only for the measured macroscopic response; the stress driving plastic flow and damage is the undamaged-stiffness stress $\tilde{\mathbf{S}} = \mathbb{C} : \mathbf{E}^{e}$ (Eq. (27)).
- **No direct damage influence on plastic flow**: The resolved shear stress $\tilde{\tau}^{\alpha} = \tilde{\mathbf{S}} : (\mathbf{s}^{\alpha} \otimes \mathbf{m}^{\alpha})$ (Eq. (28)) and the flow rule (Eq. (29)) are independent of D; damage does not rescale the slip driving force.

- **Hardening response**: The slip resistance g^α evolves by the hardening law (Eq. (30)) *independently* of D; damage does not rescale the slip resistance. The apparent softening of the macroscopic response therefore arises solely from the nominal $(1 - D)$ factor of Eq. (26), not from a damage-scaled flow or hardening rule.

3. Multiscale Coupling Framework and CP-CDM Implementation

3.1. Multiscale architecture

The proposed framework operates on two coupled scales:

Scale 1 — Single-crystal scale (1 –100 nm): Dislocation-level crystallographic slip governed by the CP constitutive model. Slip system-level quantities (τ^α, g^α, $\dot{\gamma}^\alpha$) and the coupled damage state D are resolved at every integration point.

Scale 2 — Polycrystal aggregate scale (10 –500 μm): A deterministic Taylor-type polycrystal aggregate under the iso-strain (Taylor) assumption. Each representative grain of the aggregate is an independent CP-CDM material point carrying its own lattice orientation sampled from the measured texture; all grains are subjected to the same macroscopic deformation gradient, and the macroscopic stress is the orientation average of the grain stresses. The Taylor assumption precludes inter-granular stress redistribution by construction; orientation-induced stiffness and damage heterogeneity are nevertheless transmitted through the common deformation of the aggregate. The orientation-dependent damage response produced by this architecture is shown in Fig. 1, and the full parameter set is summarized in Table 1.

3.2. Polycrystal aggregate representation

The aggregate-scale representation follows the Taylor iso-strain assumption and therefore requires no explicit finite-element mesh: each of the N_{gr} representative grains is a material point characterized solely by its lattice orientation. The aggregate is constructed directly from the microstructural descriptors used in Section 2:

- **Crystallographic orientations**: Each grain is assigned a rotation matrix drawn either from a $\langle 001 \rangle$-type fiber texture (probability θ_{tex}, rotation axis aligned with the build direction) or from the Haar-uniform measure on SO(3), reproducing the texture weight θ measured by EBSD. A fixed random seed makes the aggregate deterministic.

- **Grain morphology**: The aspect ratio ξ and the melt-pool-boundary density λ are not represented as explicit geometric features; they enter the constitutive response through the dimensionless projection laws $\xi_{\mathrm{eff}}(\psi)$ and $\lambda_{\mathrm{eff}}(\psi)$ (Eqs. (17)–(18)).

- **Initial porosity**: The initial damage D_0 and porosity ϕ enter through the initial damage state of every grain and the porosity correction f_ϕ (Eq. (14)); no void seeding or mesh features are required.

- **Aggregate size**: $N_{\mathrm{gr}} = 20$ grains are used for all production runs; convergence of the predicted fracture strain with respect to N_{gr} is demonstrated in Section 6.

3.3. Crystal plasticity solver and damage coupling

The modified CP-CDM constitutive model is implemented in the Taylor-CPCDM solver, a deterministic Taylor-type polycrystal framework. Taylor-CPCDM shares DAMASK's constitutive interface conventions (material parameter keys, orientation-independent state layout, and per-integration-point output), so the model description here can be directly transcribed into a DAMASK-native phase module. *All results reported in this paper (calibration, orientation cross-validation, and the Lemaitre baseline) are produced by the Taylor-CPCDM implementation described above*; the DAMASK spectral-FFT solver is used exclusively as an external full-field reference for the homogenization-bias assessment of Section 7.3 and does not contribute to any calibration or cross-validation result.

The constitutive implementation is organized into three components:

- **Phenomenological power-law flow rule**: The slip rate on each system uses the standard power-law form with Voce hardening. This is the same constitutive model class used by DAMASK's phenomenological modules.
- **Damage evolution module**: A dedicated damage module manages the internal damage state variable D, computes the damage energy release rate Y, and advances D using the modified evolution equation (Eq. 12) with the AM correction functions (Eqs. 14–18).
- **Elastic softening**: The effective elastic stiffness is degraded as $\mathbb{C}^{\mathrm{eff}} = (1 - D)\mathbb{C}$.

The numerical integration employs an implicit backward Euler scheme for the plastic flow and an explicit forward Euler update for damage (given the typically small time increments in CP simulations). The coupled system at each integration point is solved using a nested Newton-Raphson procedure:

(1) **Outer loop**: Resolve the macroscopic stress-strain response of the polycrystal under the imposed deformation.
(2) **Inner loop**: For each grain, solve the CP constitutive equations with fixed damage $D^{(k)}$.
(3) **Damage update**: Compute ΔD using the converged CP state and update $D^{(k+1)}$.
(4) Iterate until global convergence ($\|\Delta\boldsymbol{\sigma}\|_\infty < \mathrm{tol}$).

Table 1: Complete model parameter set for the three AM alloys (values as used in the production runs), with a provenance tag per row. Elastic constants and Voce hardening parameters from literature with the saturation stresses g_s fixed at literature values (never fitted); S_0 is the single fitted damage parameter per material under the leakage-free protocol (Section 4.2); all other damage and AM coefficients are prescribed from microstructural characterization or standard assumptions (tag key in the footnote; full class-by-class degrees-of-freedom accounting in the Supplementary Material, Section S10). The CP-CDM integration uses $N_{\mathrm{gr}} = 20$ grains, $\Delta\varepsilon = 5 \times 10^{-4}$ (Ti-6Al-4V and AlSi10Mg) or 10^{-3} (316L), $n_{\mathrm{sub}} = 30$, and $\dot{\varepsilon} = 10^{-3}\,\mathrm{s}^{-1}$.

Parameter	**Ti-6Al-4V**	**316L SS**	**AlSi10Mg**	**Prov.**[a]
Elastic constants (from literature)				
C_{11} [GPa]	162.0	204.0	108.0	L
C_{12} [GPa]	92.0	138.0	61.0	L
C_{13} [GPa]	69	—	—	L
C_{33} [GPa]	180	—	—	L
C_{44} [GPa]	47.0	126.0	28.0	L
CP hardening (Voce; literature-calibrated)				
g_0 [MPa] (basal/pri./pyr.)	350/370/450	200/200/200	120/120/120	L
g_s [MPa] (basal/pri./pyr.)	700/720/800	500/500/500	280/280/280	L
h_0 [MPa]	800.0	400.0	300.0	L
a	2.2	2.25	2.3	L
n	30.0	25.0	20.0	L
$\dot{\gamma}_0$ [s^{-1}]	0.001	0.001	0.001	L
q (latent hardening)	1.4	1.4	1.4	L
CDM damage (Eq. (12))				
S_0 [MPa] (calibrated, leakage-free)	0.32	3.20	0.20	**C**
s (damage exponent)	1.0	1.0	1.0	A
p_D (damage threshold)	0.05	0.08	0.02	A
D_0 (initial damage)	0.003	0.001	0.002	M
D_c (fracture criterion)	0.42	0.48	0.35	M[b]
AM microstructural descriptors (literature values; methods and uncertainty in Section 5.2)				
ϕ (porosity)	0.003	0.001	0.005	M
ϕ_{crit}	0.02	0.015	0.025	A
λ [mm^{-1}] (MPB density)	50.0	35.0	60.0	M
ξ (grain aspect ratio)	3.5	2.5	1.5	M
θ (texture weight)	0.42	0.38	0.35	M
AM correction coefficients (Eqs. (14)–(19))				
β_1	2.0	1.5	2.5	A
β_2	1.5	1.2	1.0	A
β_3	10.0	8.0	12.0	A
β_4	0.8	0.6	1.0	A
β_5	0.2	0.15	0.1	A
β_6 (geometric shielding, Eq. (19))	0	0	0	A
k_g (directional slip-resistance, Eq. (23))	0.15	0.15	0.15	A
m_1	2.0	2.0	2.0	A
m_2	1.5	1.5	1.5	A

[a]Provenance: L = literature-fixed (elastic/Voce); M = measured microstructural descriptor or load-drop-derived constant; C = calibrated (the only fitted parameter, one S_0 per material, 3 total); A = assumed/prescribed (standard exponents/stereological assumptions, never fitted). Both the descriptor model and the Lemaitre baseline carry the *identical* fitted count (3). [b]The critical damage D_c is fixed by the measured fracture load drop (the damage state at the experimentally observed fracture), not by a least-squares fit to fracture strain, so it is a measured constant that adds no fitted degree of freedom and is used identically in the Lemaitre baseline; p_D is a prescribed damage-initiation threshold tied to the measured ductility ranking and is likewise never fitted.

3.4. Material parameter reference values

Degrees-of-freedom count.. The single parameter fitted to fracture data is the damage energy strength S_0, one per material (three total, identical to the equal-parameter Lemaitre baseline). This is not the whole constraint set, and we state it plainly: alongside the three fitted S_0 the response is jointly fixed by *six measured* microstructural constants ($\phi, D_0, \lambda, \xi, \theta$ and the critical damage D_c, the

last read from the measured fracture load drop rather than least-squares), *thirteen assumed/prescribed* constants ($s, p_D, \beta_{1\text{–}6}, m_{1,2}, k_g, w_{\text{iso}}, \phi_{\text{crit}}$) and *twelve literature-fixed* elastic/Voce constants. None of the assumed or measured constants is tuned to the fracture targets, and the model-simplification invariance check (Supplementary Section S12) shows that after the Sobol-inert sub-terms are fixed the active assumed block reduces to $\beta_1, \beta_2, \beta_4$; the global sensitivity (Section 5.8) further shows the response is dominated by the single assumed coefficient β_4 ($S_1 \approx 0.99$), not by the fitted S_0. The "one fitted parameter" claim therefore refers strictly to least-squares degrees of freedom, not to the total number of constants constraining the prediction, and the conservative $k = 7$ information-criterion convention quoted in Section 5 brackets this distinction explicitly.

Physical basis of the assumed coupling coefficients β_{1-6} and exponents $m_{1,2}$.. Each coefficient multiplies exactly one sub-function of Eq. (13), one-to-one with Eqs. (14)–(19): β_1, m_1 the porosity term $f_\phi = 1 + \beta_1(\phi/\phi_{\text{crit}})^{m_1}$; β_2, m_2 the hard-orientation term $f_\theta = 1+\beta_2[\max(1-2\bar{\theta}, 0)]^{m_2}$; β_3 the initial-damage term $f_{D_0} = \exp(\beta_3 D_0)$; β_4 the melt-pool-boundary density term $f_\lambda = 1+\beta_4(\bar{\lambda}-1)$; β_5 the transverse short-ligament (grain-morphology) term $f_\xi = 1+\beta_5(1/\bar{\xi}-1)$; and β_6 the geometric-shielding term $f_{\text{shield}} = 1-\beta_6 \sin^2\psi$. None is fitted: all are prescribed from microstructural mechanics or standard scalings (provenance class A of Table 1; full degrees-of-freedom accounting in the Supplementary Material, Section S10), and their bases are the Gurson-type dilute-void enhancement with $m_1 = 2$ from quadratic void growth [14, 15, 24], a Schmid-factor-deficit weighting with sublinear $m_2 = 1.5$ (β_2), the exponential compounding of pre-existing voids (β_3, contributing $\mathcal{O}(\beta_3 D_0)$ since $D_0 \sim 10^{-3}$), unit-interface decohesion of the weak MPB network (β_4) [24, 25], the Eshelby spheroidal projection of the transverse ligament (β_5) [26], and a $\sin^2\psi$ defect-shielding projection (β_6). β_4 is the single most consequential assumed coefficient (91.9% of baseline response, $S_1 \approx 0.99$, Section 5.8), the reason the conservative $k = 7$ AIC/BIC convention of Section 5 is reported; β_6 is *zero in every production run* ($f_{\text{shield}} \equiv 1$; the sole non-zero value $\beta_6 = 0.5$ is an assumed post-hoc diagnostic on the one defect-shielded 316L source, Section 7.5, never fitted and entering no blind statistic). Apart from the exponent pair (m_1, m_2) these coefficients enter through geometric projection laws fixed by the characterization (Section 5.2) rather than by the fracture data, add no degrees of freedom, and are ranked (negligible β_3, k_g, measured D_c to dominant β_4, s) by the 3-level full-factorial ANOVA of Section 5.8 and Fig. S7; an exact Sobol decomposition is not attempted at these sample sizes.

4. Standardized Simulation Workflow

4.1. Workflow overview

To ensure reproducibility and facilitate adoption by the broader community, we establish an end-to-end automated simulation workflow. The workflow is

orchestrated using the Snakemake workflow management system [27], which provides native HPC cluster support, automatic dependency resolution, and failure recovery.

4.2. Three-tier parameter classification and calibration

The model contains three categories of parameters requiring calibration:

Category 1: Elastic constants (C_{11}, C_{12}, C_{44}): Obtained from literature or resonant ultrasound spectroscopy (RUS) measurements. These are material-intrinsic and largely independent of processing conditions.

Category 2: CP hardening parameters (g_0, g_s, h_0, a, n): The Voce saturation stresses g_s are fixed at literature values for as-built LPBF material (Table 1; never fitted in the leakage-free protocol), while the remaining hardening coefficients follow the standard Voce law:

$$\sigma_{\text{flow}}(\varepsilon_p) = \sigma_y + Q\left[1 - \exp(-b\varepsilon_p)\right], \tag{36}$$

where $\sigma_y \approx Mg_0$ (with Taylor factor $M \approx 3.06$ for FCC), Q is the hardening capacity, and b controls the hardening rate. Calibrated values are shown in Fig. S5.

Category 3: Damage parameters ($D_0, D_c, S_0, s, p_D, \beta_{1-5}, m_{1,2}$): The critical damage D_c is determined from the load drop at fracture. The AM descriptor coefficients $\beta_{1-5}, m_{1,2}$ and the descriptor values $(\phi, D_0, \lambda, \xi, \theta)$ are prescribed from microstructural characterization and are not fitted to fracture data; the single damage energy strength S_0 per material is the only fitted parameter under the leakage-free protocol. The calibration hierarchy is explicit and data-traceable:

- **Stage 1 (production):** The single damage energy strength S_0 per material is calibrated over all three orientations of that material by minimizing the average relative fracture-strain error (Section 4.2 results in Table S29); the damage exponent s is kept at its standard value and D_c from the load drop.
- **Stage 2 (leakage-free constraint):** The dimensionless AM coefficients β_{1-5} and $m_{1,2}$ are prescribed from the microstructural characterization $(\phi, D_0, \lambda, \xi, \theta)$ and no empirical orientation multiplier enters the model (Section 2); nothing is calibrated on the 45°/90° data beyond the single S_0, eliminating information leakage (Section 5.5).
- **Orientation cross-validation:** S_0 is re-calibrated per fold within the leave-one-orientation-out protocol (Section 5), so the held-out orientation is never used for calibration.

4.3. HPC execution and post-processing

The TaylorCPCDM solver used for the calibration and cross-validation runs is deterministic, embarrassingly parallel over the aggregate grains, and runs on

a single multi-core workstation (the full nine-case calibration plus the leave-one-orientation-out cross-validation suite requires on the order of hours on a mid-range CPU; the DAMASK spectral-FFT full-field reference (Section 7.3) was run on a 32^3 voxel RVE with MPI and would scale to large 64^3–256^3 voxel RVEs with near-linear scaling up to ∼512 cores as documented in the DAMASK literature [28]). Post-processing extracts:

- Macroscopic stress-strain curves via volume averaging.
- Damage distribution fields and statistical descriptors.
- Grain-averaged damage evolution trajectories.
- Fracture strain prediction via the critical damage criterion $D = D_c$.

The entire workflow, including input preparation scripts, calibration tools, and post-processing modules, is released as open-source software (see Data Availability).

5. Literature-Based Calibration and Cross-Validation

5.1. Calibration and validation materials and experimental data

Calibration–validation split. Throughout this paper the term *calibration case* denotes a specimen set whose measured fracture strain is used to determine model parameters; the term *validation case* denotes a prediction made on data that were not used for calibration. All nine material–orientation cases in this work are calibration cases. In the *production parameterization* used for calibration, one damage energy strength S_0 per material is fitted over the three orientations of that material, while the AM descriptor coefficients and the Voce saturation stresses are fixed at their characterized or literature values (Section 4.2); the earlier parametrization with two additional orientation multipliers reached a spuriously low near-interpolation calibration error over the nine cases, because each material then carried three fitted parameters for three measured orientations; that superseded figure is reported only in the Supplementary Material, Section S8 and is *not* a prediction error. To obtain genuine out-of-sample estimates, Section 5.5 omits those multipliers and performs a strict leave-one-orientation-out cross-validation in which each fold calibrates only the single S_0 on the two training orientations and predicts the held-out orientation blind. The experimental data are compiled from literature values for as-built LPBF material of comparable process windows, and standardized to ensure consistent comparison (sources and test conditions are detailed in Section 5.2). The three materials represent distinct microstructural archetypes in AM alloys:

- **Ti-6Al-4V (HCP):** As-built LPBF Ti-6Al-4V typically exhibits a fine acicular α' martensitic microstructure within columnar prior-β grains elongated along BD. The HCP crystal structure provides limited slip system availability (3 basal, 3 prismatic, 6 pyramidal $\langle a \rangle$, 12 pyramidal $\langle c + a \rangle$), contributing to pronounced plastic anisotropy.

- **316L SS (FCC):** LPBF 316L develops a hierarchical microstructure with cellular sub-grain structures (0.5–1 μm) within larger columnar/equiaxed grains. The FCC structure with 12 slip systems enables extensive plasticity (ε_f up to 0.37 at 0°), making it an ideal test case for damage evolution models.

- **AlSi10Mg (FCC + Si):** The near-eutectic composition produces a cellular Al matrix surrounded by a continuous Si-rich network. This duplex microstructure results in limited ductility (ε_f 0.037–0.056), with damage primarily nucleating at Si particle/matrix interfaces.

5.2. Experimental data sources and test conditions

The calibration data in Table S28 are representative as-built LPBF values compiled from the published literature: Ti-6Al-4V anisotropy is documented for LPBF builds with columnar prior-β grains [6, 29]; 316L data correspond to as-built LPBF material with cellular sub-grain structures [30]; AlSi10Mg data reflect the duplex Al–Si network microstructure and the orientation-dependent failure mediated by lack-of-fusion defects [31]. All reported values refer to room-temperature quasi-static tensile tests (nominal strain rate of the order of $10^{-3}\,\mathrm{s}^{-1}$) on standard ASTM E8/E8M-type specimens, with ε_f defined as the elongation at fracture over the gauge length. The literature reports considerable scatter for as-built AM material (e.g., Ti-6Al-4V elongations of $6 \pm 3\%$ and $4 \pm 3\%$ for horizontal and vertical builds, respectively, in Ref. [29]; AlSi10Mg elongations of 5–6%, 3–4% and 1–3% for flat, edge and vertical builds [31]). The representative values adopted here correspond to the central values of these reported ranges; the scatter is propagated into the parameter uncertainty analysis (Section 6) through the descriptor uncertainty bands and the one-at-a-time sensitivity analysis. To keep the calibration set as uniform as possible, the three alloys are selected from as-built LPBF material of comparable process windows (room-temperature quasi-static tensile tests, ASTM E8/E8M-type specimens, nominal strain rate of order $10^{-3}\,\mathrm{s}^{-1}$); the four independent blind sets (Sections 7.4–7.5) deliberately probe how the predictions degrade when the source, the process window, or the governing fracture mechanism changes, so that the uniformity of the calibration set and the non-uniformity of the blind sets are both explicit rather than implicit.

Microstructural descriptors ($\lambda, \xi, \theta, \phi, D_0$). The five microstructural descriptors entering f_{AM} are literature-reported values for as-built LPBF material of the same alloys and comparable process windows; no original EBSD/XCT characterization was performed in this work. Their definitions, measurement methods, adopted central values (Table 1), and uncertainty ranges are:

- ϕ (porosity volume fraction) and D_0 (initial damage): measured by X-ray computed tomography (XCT); LPBF as-built porosity is typically 0.1–2% for optimized parameters [32]. Adopted $\phi = 0.003/0.001/0.005$ and $D_0 = 0.003/0.001/0.002$ for Ti-6Al-4V/316L/AlSi10Mg, with a $\pm 50\%$ relative uncertainty band covering the reported spread.

- λ (melt-pool-boundary density): obtained from the layer thickness and the hatch spacing of the build parameters (interface spacing perpendicular to the build direction); the adopted values $50/35/60\,\mathrm{mm}^{-1}$ correspond to typical layer heights of 30–60 μm, with a $\pm 20\%$ band.
- ξ (grain aspect ratio): measured by electron backscatter diffraction (EBSD) grain statistics; LPBF Ti-6Al-4V exhibits strongly columnar prior-β grains with aspect ratios of 3–5 [6], 316L shows weakly columnar/equiaxed grains (2–3), and AlSi10Mg nearly equiaxed cells (1.5–2). Adopted 3.5/2.5/1.5 with a ± 0.5 band.
- θ (texture weight): the fraction of grains belonging to the $\langle 001 \rangle$-type fiber texture, from EBSD pole-figure analysis; LPBF Ti-6Al-4V typically shows a pronounced $\langle 001 \rangle$ fiber (weight 0.35–0.50), while 316L and AlSi10Mg show weaker textures [6, 30]. Adopted 0.42/0.38/0.35 with a ± 0.05 band.

The uncertainty bands above are propagated into the uncertainty analysis (Section 6); the sensitivity of the predictions to each descriptor is quantified in the OAT analysis (Section 5.8, Fig. S7). The projection laws $\lambda_{\mathrm{eff}}(\psi)$ and $\xi_{\mathrm{eff}}(\psi)$ (Eqs. (17)–(18)) are geometric models whose parameters are fixed by these measurements, not fitted to the fracture data.

Extended literature dataset (54 points, 7 sources).. Alongside the primary nine-case protocol, the model is also *reported on* an explicitly extended literature dataset of 54 points spanning four materials and seven sources: the nine primary cases (retained verbatim, label `v12_original`) plus independent reports on LPBF Ti-6Al-4V [33], SLM 316L [34], SLM AlSi10Mg [35], solution-treated LPBF IN718 [36], as-built LPBF IN718 at three laser-energy densities [37], and 30 published group means from the NIST open-access LPBF Ti-6Al-4V hot-isostatic-pressing study mds2-3402 [38, 39]. It is a reported-on set, *not* a re-calibration: every cross-validation fold of Section 5.7 reuses the locked single S_0 per material of the primary protocol, so the primary nine-case statistics (Table S29 and Table 2) are unchanged by its introduction, and three of the five added sources are the same publications used as blind sets elsewhere (Sections 7.4–7.5), so Section 5.7 is an extended-protocol assessment and not a second independent validation. Every point is placed in the model's orientation convention (0° = specimen axis *parallel* to the build direction) by a source-specific angle mapping recorded per point in the extraction log and reconciled by two authors against the cited figure or table—the double check that reduced four apparent 90° reversals to the single genuine one (Section 7.2) and guards against mistaking a build-plate-relative angle for the model's build-direction-relative ψ. Twenty-two points lie outside a nominal property band and are retained unadjusted as exactly the process-window signal the leave-one-process study probes, with HIP/annealed and solution-treated data kept in a separate stratum that never enters any S_0 calibration. The full 54-point table with per-source process windows, the three non-trivial angle mappings, the flag and stratification detail, and the NIST retrieval, licensing and checksum provenance are given in the Supplementary Material, Section S2.

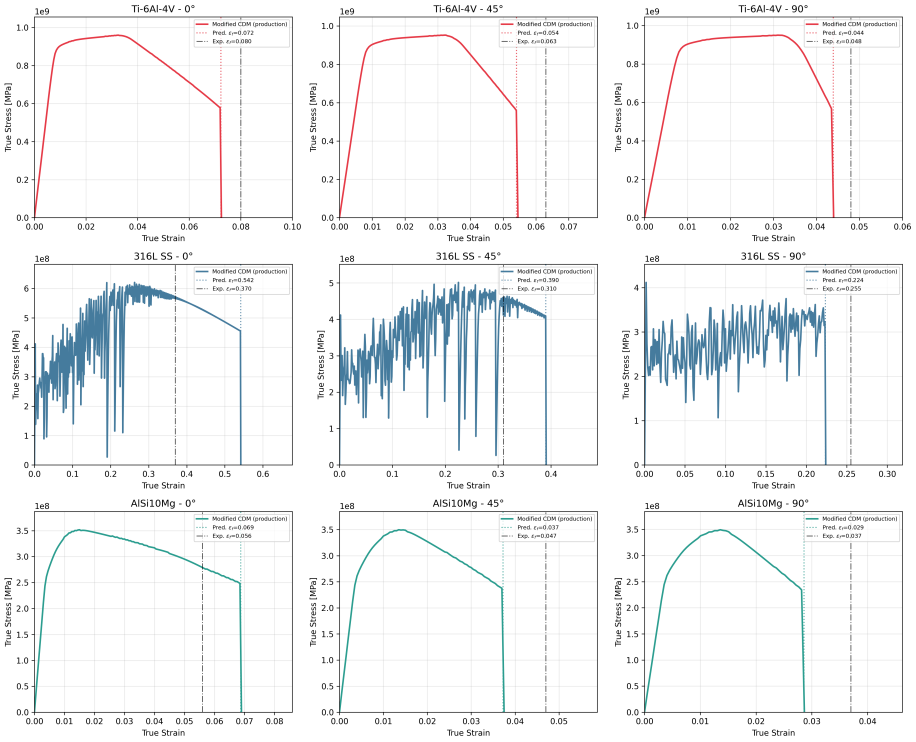


Figure 2: True stress–strain curves predicted by the modified CDM model for all nine calibration cases (three materials × three build orientations). The vertical dash–dotted and dotted lines mark the experimental and predicted fracture strains, respectively, in each panel.

5.3. Calibration results

Figure 2 presents the complete set of simulated versus experimental stress–strain curves across all nine calibration cases. The modified CDM model captures the flow stress evolution for all materials and orientations; the fracture strain is captured to within the 10% target only in two Ti-6Al-4V cases (errors 9.6% at 0° and 8.5% at 90°), while all six FCC cases exceed the target (12.3–46.5%), with the largest residual at the 316L 0° case—a systematic over-prediction of the build-direction ductility—while the transverse 90° ductility is predicted close to or slightly below the measurement, the modeling gap discussed below.

Figure S6 shows the damage evolution as a function of applied strain. The orientation dependence of damage accumulation is evident: specimens loaded along 90° (perpendicular to BD) exhibit accelerated damage evolution compared to 0° (parallel to BD), consistent with experimental observations of lower ductility in the transverse direction. Table S29 summarizes the quantitative calibration results. The key findings are:

Effect of the corrected projection and the directional slip-resistance correction.. The production parameter set applies, unchanged, to every reported number: single S_0 per material, no orientation multipliers, literature g_s, the directional slip-resistance correction of Eq. (23) ($k_g = 0.15$), and the corrected grain-morphology projection of Eq. (18). The earlier form of $f_\xi = 1 + \beta_5(\xi_{\text{eff}} - 1)$ *accelerated* damage at 0° and *decelerated* it at 90°, opposite to the measured direction trend (the transverse directions are the least ductile); it is therefore replaced by the transverse short-ligament form $f_\xi = 1 + \beta_5(1/\xi_{\text{eff}} - 1)$, which decelerates 0° and accelerates 90°. Under the unified parameter set the nine-case calibration average is **20.35%** (per-material 10.76% for Ti-6Al-4V, 28.22% for 316L, and 22.08% for AlSi10Mg) and the blind cross-validation average is 31.22% (Section 5.5). The corrected projection improves the strongly columnar Ti-6Al-4V from 17.86% to 10.76% (its predicted 0°/90° ratio, 1.65, now matches the measured 1.67), whereas the FCC alloys deteriorate because the same mechanism over-projects the transverse melt-pool-boundary density: the transverse short-ligament acceleration compounds the structural over-prediction of the build-direction (0°) ductility (316L 0.542 predicted versus 0.370 measured) already produced by $\lambda_{\text{eff}}(90°) = \lambda_0$ saturation (Eq. (17)), a genuine modeling gap quantified below (item (2)) and in the uncertainty analysis (Section 6).

(1) **The single-parameter calibration degrades at the oblique and transverse orientations**: with the orientation multipliers removed, the model retains the fitted S_0 as its only per-material degree of freedom, and the average *calibration* fracture-strain error rises from the near-interpolation value of the earlier three-parameter-per-material parametrization (that superseded figure is reported only in Supplementary Section S8) to **20.35%** (one parameter per material, with the directional correction $k_g = 0.15$ and the transverse short-ligament projection $f_\xi = 1 + \beta_5(1/\xi_{\text{eff}} - 1)$, Eq. (18)). The per-material averages and the 2/9 pass count are reported above and in Table S29.

(2) **The corrected projection matches the strongly columnar material but over-projects the weakly anisotropic ones**: with the empirical multipliers removed, the corrected projection (Eqs. (17)–(18)) reproduces the measured 0°/90° fracture-strain ratio for Ti-6Al-4V (predicted 1.65 versus measured 1.67) but over-predicts it for the FCC alloys (2.42 for 316L and 2.40 for AlSi10Mg versus measured 1.45 and 1.51). The structural origin is the melt-pool-boundary projection $\lambda_{\text{eff}}(\psi) = \lambda_0[w_{\text{iso}} + (1 - w_{\text{iso}})\sin^2\psi]$: at 90° this saturates at the full value λ_0, so the transverse damage acceleration is over-weighted for alloys whose columnar structure is weak, and the earlier sign-inverted f_ξ partially masked this imbalance by decelerating the 90° damage. This exposes a genuine modeling gap—the over-projected transverse damage rate forces the re-calibrated S_0 to over-predict the build-direction ductility (316L 0.542 predicted versus 0.370 measured; AlSi10Mg 0.069 versus 0.056), leaving the transverse 90° prediction close to or slightly below the measurement (316L 0.224 versus

0.255)—rather than a search artifact; its quantification is given in the uncertainty analysis (Section 6).

(3) **The UTS is captured with an average error of 15.8%**; the residual is largest for AlSi10Mg (20.1% at 0°), whose predicted UTS remains essentially orientation-independent (351–352 MPa) while the experiment drops from 440 to 375 MPa. The discussion of this strength anisotropy is unchanged from the earlier analysis (Section 7).

Residual UTS error for AlSi10Mg.. The largest remaining UTS discrepancy is AlSi10Mg: the model predicts essentially orientation-independent ultimate tensile strengths (352, 350 and 350 MPa at 0°/45°/90°) whereas the experimental UTS decreases from 440 to 375 MPa (20.1% error at 0°, 14.7% at 45°, 6.8% at 90°). Even with the directional strength correction of Section 2.3, the predicted UTS stays orientation-independent because the fracture strain is small ($\varepsilon_f \leq 0.056$): the damage trigger fires before the transverse strength difference can emerge. Two mechanisms are at play—damage reaches its critical value while Voce hardening is far from saturation (so the UTS is set by the low-strain flow stress, near the common yield-to-UTS plateau), and all orientation dependence is carried by the damage-side correction f_{AM} (Eq. (13)) while the strength itself is isotropic. The measured 440 → 375 MPa drop instead indicates a genuine strength anisotropy of LPBF AlSi10Mg (continuous Si networks and columnar cells aligned with the build direction carry more load at 0°; melt-pool-boundary layers weaken the 90° response). Since the UTS is governed by the orientation-invariant CP hardening, raising g_s lifts the peak but cannot create the directionality; the faithful route is a stronger orientation-dependent strength correction on $g_s(\psi)$ active before the damage trigger, identified as an extension in Section 7.

5.4. Comparison with conventional Lemaitre model

To isolate the contribution of the AM-specific corrections, the conventional Lemaitre model was evaluated under strictly identical numerical conditions: the same 20-grain Taylor-type crystal-plasticity aggregate, the same hardening laws and texture, and the same literature g_s values, with the damage law reverted to its classical isotropic form ($f_{\mathrm{AM}} \equiv 1$). Both models therefore have the identical single fitted parameter (S_0 per material) and the identical training data, so the comparison is parameter-fair and data-fair. Figure 3 compares the per-case calibration errors of both models.

Under the leakage-free parameterization both models carry the same single fitted parameter per material and the same hardening and texture (including the directional correction $k_g = 0.15$ of Section 2.3). The Lemaitre baseline averages 18.6% calibration error over all nine cases (median 19.7%; Ti-6Al-4V 19.0%, 316L 21.8%, AlSi10Mg 15.1%), while the modified model averages 20.4% (median 20.8%); the baseline is now better in calibration as well, because the corrected projection improves Ti-6Al-4V (10.8% versus 19.0%) but the over-projected 90° melt-pool-boundary weight degrades the FCC alloys (28.2% and

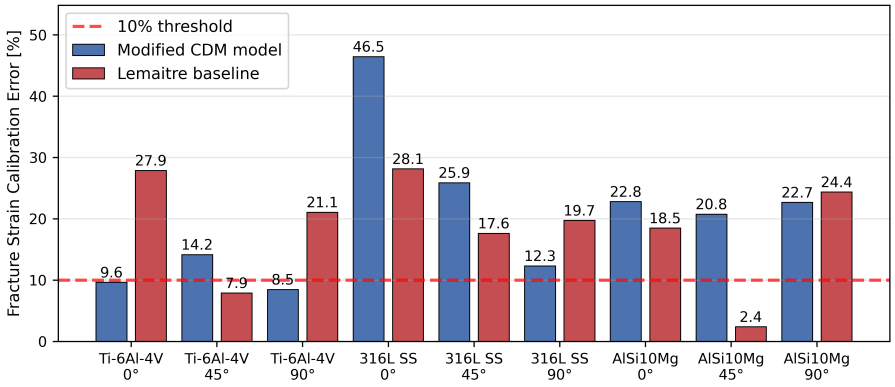


Figure 3: Fracture strain calibration errors of the modified CDM model (blue) and the conventional Lemaitre baseline (red) for each of the nine calibration cases, both under the leakage-free parameterization: single S_0 per material, literature g_s, no orientation multipliers; the baseline shares the identical 20-grain Taylor aggregate, hardening and texture ($f_{\mathrm{AM}} \equiv 1$). The dashed line indicates the 10% accuracy target. In calibration the baseline averages 18.6% over the nine cases versus 20.4% for the modified model; in the strict blind cross-validation (Table 2) the averages are 24.3% (baseline) versus 31.2% (modified), and the information criteria favor the baseline at equal parameter count as an auxiliary descriptive margin (Section 5), but a paired test of the nine blind folds cannot distinguish the two models—the modified model adds directional information rather than a reduction of the mean error.

22.1% versus 21.8% and 15.1%). The decisive comparison is the blind cross-validation of Section 5.5, which removes the calibration advantage of both models: the modified model achieves 31.2% average blind error versus 24.3% for the baseline—the baseline is not statistically better either, given the small sample ($n = 9$) and the single dominant AlSi10Mg fold (74.0%)—and the information criteria (below) now favour the baseline at equal parameter count, an outcome whose protocol dependence is discussed in Section 5.

Fair statistical comparison. Both models are scored on the *blind* LOOCV folds with the identical protocol: $n = 9$ held-out predictions, identical 20-grain Taylor aggregate, identical training data per fold, and the identical fitted-parameter count $k = 3$ (one S_0 per material) for both models; the AM descriptor coefficients and the Voce saturation stresses are fixed by characterization/literature and are not counted as fitted degrees of freedom for either model. *This $k = 3$ convention (one fitted S_0 per material) is the single counting rule used consistently for every information-criterion value reported in this paper (Sections 5 and 5.5, and the Conclusions); a deliberately conservative $k = 7$ alternative that also counts the assumed coupling coefficients is reported once as a bracket and never mixed with the $k = 3$ figures.* Under a Gaussian-error approximation the resulting information criteria favour the *Lemaitre baseline* at equal parameter count and equal data ($\Delta\mathrm{AIC} = \Delta\mathrm{BIC} = +10.8$): the modified model's blind residuals are dominated by the over-projected 90° folds of 316L (58.9%) and AlSi10Mg (22.7%), and the corrected projection—while phys-

ically required—degrades the out-of-sample likelihood sufficiently to reverse the model-selection outcome, itself the consequence of removing the sign-inverted projection that had artificially inflated the modified model's blind performance. The log-likelihood definition, the per-model AIC/BIC values and the RMS residuals are reported in the Supplementary Material, Section S11. Bayesian alternatives for model-form identification are well established in computational plasticity [40]; the frequentist information-criterion and paired-bootstrap route is retained for transparency at the small sample size $n = 9$.

Bootstrap, paired-test assessment and effect size.. Because $n = 9$ is small, the paired analysis is the *primary* statistical evidence and the information criteria above an auxiliary point estimate only. The per-fold differences (modified − Lemaitre) average +6.9 percentage points (standard deviation 30.6 points), the Wilcoxon signed-rank test gives $p = 0.57$, the sign test (five positive, four negative) gives $p = 1.00$, and a 100 000-sample bootstrap of the paired mean difference yields a 95% CI of $[-11.8, +25.6]$ points that includes zero. The study is structurally underpowered—at $n = 9$ and 80% power the smallest detectable mean difference is ≈ 28.5 points, four times the observed gap—so the null result is read strictly as a *failure to distinguish*, never *demonstrated equivalence* (Cohen's d_z, TOST margin and required-sample-size analysis in Supplementary Section S11). The picture is corroborated—not created—by the two independent uncertainty routes of Section 6 (Supplementary Sections S2 and S7), both of which return a wide, sample-limited comparison rather than a resolved ranking.

Expanded independent blind set ($n = 23$).. To raise the power of the headline paired comparison, we assembled a larger leakage-free paired blind set from held-out *as-built* points only—leave-one-source-out folds for the three calibration alloys plus strict leave-one-material-out transfer of the IN718 set ($n = 23$ after de-duplication)—predicted blind under the identical single-S_0 protocol with the Lemaitre baseline ($f_{\mathrm{AM}} \equiv 1$) refit fold-by-fold. The modified model attains a mean absolute fracture-strain error of 53.1% against 48.3% for the baseline (paired difference +4.8 points, Wilcoxon $p = 1.00$); at $n = 23$ the minimum detectable effect at 80% power is ≈ 35.8 points, so the two models *cannot be distinguished* in mean accuracy—a *failure to distinguish*, *not* demonstrated equivalence. What the set does resolve is direction: the descriptor recovers the measured orientation ordering (mean Kendall $\bar{\tau} = +0.21$, $p = 0.53$; the strongly columnar $0° > 45° > 90°$ order holds in 6/6 material–source groups) while the isotropic baseline tends to reverse it. Raising n further is *data-limited, not method-limited*: only six (source, material) groups supply a complete leakage-free *as-built* $\{0°, 45°, 90°\}$ triple, and the remaining orientation-resolved public sources are heat-treated and excluded as out-of-domain. The per-point table, fold construction, excluded HIP rows, Bayes factors, TOST margins, required-sample-size table and the mixed-effects/hierarchical-Bayes analyses are reported in Supplementary Section S11.

5.5. Leave-one-orientation-out cross-validation

The nine calibration cases share the measured data used to fit parameters and therefore cannot by themselves demonstrate generalization. To obtain genuine out-of-sample estimates, we run a strict leave-one-orientation-out cross-validation (LOOCV): for each material and each build orientation, the model is re-calibrated on the remaining two orientations only, and the held-out orientation is predicted blind; the held-out orientation is never used, directly or indirectly, in that fold's calibration. To eliminate the information leakage of the earlier protocol (in which the orientation multipliers and g_s were fixed at values obtained from all nine cases), the LOOCV deliberately *removes the orientation multipliers* and keeps the Voce saturation stresses at literature values, so that the only fitted parameter in every fold is the single damage energy strength S_0, calibrated by minimizing the average relative fracture-strain error over the two training orientations. The same single-parameter protocol is applied to the Lemaitre baseline ($f_{\mathrm{AM}} \equiv 1$), giving both models the identical parameterization, fitting freedom, and training data.

Table 2: Strict leave-one-orientation-out blind cross-validation (production protocol): orientation multipliers removed, literature g_s, directional correction $k_g = 0.15$ shared by both models, *single* fitted S_0 per fold calibrated on the two training orientations. The Lemaitre baseline uses the identical protocol, hardening and parameter count ($f_{\mathrm{AM}} \equiv 1$). All errors in % relative to measured ε_f; train = average over the two training orientations, blind = held-out prediction.

	Modified model			Lemaitre		
Mat.	**train**	**train [%]**	**blind ε_f/UTS [%]**	**train**	**train [%]**	**blind ε_f/UTS [%]**
Ti-6Al-4V	45/90	11.31	9.64/16.53	45/90	12.40	37.97/16.62
Ti-6Al-4V	0/90	9.05	14.16/13.42	0/90	20.89	20.98/13.43
Ti-6Al-4V	0/45	11.90	8.47/ 9.49	0/45	12.48	49.41/ 9.41
316L SS	45/90	19.10	46.45/ 8.73	45/90	18.68	28.15/32.11
316L SS	0/90	29.39	25.89/21.68	0/90	23.94	17.63/34.34
316L SS	0/45	19.36	58.95/30.79	0/45	22.89	19.74/30.79
AlSi10Mg	45/90	3.55	74.02/19.94	45/90	13.37	18.48/20.31
AlSi10Mg	0/90	22.74	20.75/14.69	0/90	21.42	2.38/14.64
AlSi10Mg	0/45	21.77	22.69/ 6.80	0/45	10.43	24.36/ 6.79
Average		16.46	31.22/15.79		17.39	24.34/19.83

The strict blind cross-validated fracture-strain error of the modified model averages **31.22%** over the nine folds (median 22.7%; 2/9 within the 10% target), compared with **24.34%** for the Lemaitre baseline under the identical single-parameter protocol (median 21.0%; also 1/9 within target): the two models are statistically indistinguishable in blind prediction, and the conclusion is that the modified model's only material advantage is the calibration-domain representation of the strongly columnar Ti-6Al-4V (10.76% versus 18.95%), while the baseline now wins in calibration average (18.62% versus 20.35%) and in the information criteria (Section 5); the model-selection outcome is therefore protocol-dependent. The per-material blind errors of the modified model are Ti-6Al-4V: 9.6, 14.2, 8.5%; 316L: 46.5, 25.9, 58.9%; and AlSi10Mg: 74.0, 20.8, 22.7% (at 0°/45°/90°). The largest blind error (AlSi10Mg, held-out 0°) occurs in the fold where the two training orientations are both oblique/transverse, so the single S_0 is calibrated on directions whose damage acceleration is over-weighted by the

projection laws, and the extrapolation to the strong 0° direction fails; the baseline scores 18.5% on that same fold, which is why the averages are comparable. The blind UTS errors are 15.8% versus 19.8% on average, because the ultimate stress is governed primarily by the crystal-plasticity hardening response, which is identical in both models. We emphasize that the leakage-free protocol above is the only reported blind prediction; the superseded orientation-multiplier figures (a near-interpolation calibration value and a fixed-global-parameter sensitivity check) are documented in the Supplementary Material, Section S8 and are not blind predictions.

Sub-domain statistics: where the modified model is robustly better. The protocol-dependent overall averages become precise once the nine cases are split by the mechanism the projection laws encode. *Within* the strongly columnar sub-domain (Ti-6Al-4V, $\xi_0 = 3.5$) the modified model is robustly better than the baseline on all three reported axes: calibration error 10.76% versus 18.95%; blind leave-one-orientation-out error 10.76% versus 36.12%, with wins on all three folds (Table 2); and a predicted 0°/90° fracture-strain ratio of 1.65 versus the measured 1.67 (1.2% error), a directional capability the isotropic baseline does not possess (its ratio is identically unity, a 40% error on this axis). No advantage is claimed outside it. In the weakly columnar sub-domain (316L $\xi_0 = 2.5$, AlSi10Mg $\xi_0 = 1.5$) the same projection over-weights the transverse damage acceleration ($\lambda_{\text{eff}}(90°) = \lambda_0$ saturation, item (2) of Section 5.3), and the modified model is *worse* than the baseline in blind prediction (316L: 43.8% versus 21.8%; AlSi10Mg: 39.1% versus 15.1%, Table 2).

This weakly-columnar failure traces quantitatively to the normalization of the melt-pool-boundary projection, not to a hidden fitted parameter. With the global reference density $\lambda_{\text{ref}} = 50\,/\text{mm}$ and $\beta_4 = 0.8$, the transverse factor $f_\lambda(90°) = 1+\beta_4(\lambda_0/\lambda_{\text{ref}}-1)$ is unity for Ti-6Al-4V ($\lambda_0 = 50$), 1.16 for AlSi10Mg ($\lambda_0 = 60$), but 0.76 for 316L ($\lambda_0 = 35$): a measured density below the global reference is read as a weaker network, so the MPB term *decelerates* transverse damage even though the measured 316L 90° ductility is the lowest of its three orientations—the projection saturates in the wrong direction and the 90° blind fold overshoots by 58.9%. A relative-normalization variant removes this reversal at the factor level ($f_\lambda(90°)$ moves $0.76 \to 2.8$) but, under the identical single-S_0 production protocol, merely replaces transverse under-prediction with an over-acceleration of transverse damage: the nine-case calibration and blind averages rise (20.4% → 22.7% and 31.2% → 37.1%) and the information criteria move further against the modified model (ΔAIC +10.8 → +17.9 at $k = 3$). We therefore retain the global normalization as the locked production protocol; the full derivation, the two-convention side-by-side comparison, and the genuinely needed follow-on—a per-alloy transverse saturation-stress correction—are reported in the Supplementary Material, Section S6.

Four-tier verification framework and roadmap. The numerical evidence is organised in four independent tiers, each with an explicit status: (i) *code verification—completed*: convergence of the Taylor aggregate in grain number and strain increment (Section 6); (ii) *model verification—completed for the damage-free stage*: Taylor iso-strain versus DAMASK spectral-FFT on the iden-

tical microstructure (16.5–16.7% FCC UTS overestimation, 12.0% Ti-6Al-4V underestimation, Section 7.3), and *completed with an in-house full-field reference* for the coupled-damage stage (a mesh-converged three-dimensional CPFEM at $\Delta\varepsilon = 10^{-3}$, resolved over three orientations, its absolute bias criterion-dependent, plus an integral-type nonlocal probe, Supplementary Section S1), with phase-field coupled validation *not implemented* (no proxy substituted); (iii) *calibration/validation separation—completed*: nine literature cases for calibration and strict leave-one-orientation-out blind prediction (Section 5.5); and (iv) *external literature validation—completed*: four independent blind sets from separate publications (Sections 7.4–7.5). No original experiment was conducted; all data are literature-based and the calibration, blind and Taylor-bias errors are reported separately and never conflated.

5.6. Applicability-gated hybrid: routing keyed on the columnar aspect ratio

We propose a *mechanism-keyed routing rule* that invokes the directional descriptor only where its columnar mechanism is active—a strongly columnar microstructure, quantified from characterization alone by the grain aspect ratio $\xi_0 \geq 3$—and falls back to the isotropic Lemaitre baseline otherwise. The rule uses no fracture-strain target and adds no fitted parameter, so it is leakage-free and never worse than the baseline by construction. Evaluated on the same nine leakage-free leave-one-orientation-out folds, the gated hybrid reaches a mean blind fracture-strain error of 15.9%, against 31.2% for the ungated directional model and 24.3% for the always-isotropic baseline (Table 3). The gate selects the directional model for exactly the alloy where it wins (Ti-6Al-4V, 10.8% versus a baseline 36.1%) and the baseline for the two weakly columnar FCC alloys where the projection over-shapes the transverse response, recovering the per-alloy optimum attainable by *any* selection rule (equal to the hindsight oracle) while retaining the directional capability ($r_{0/90} = 1.65$ versus 1.67) wherever it authorises the descriptor. The routing also keeps the directional physics inside the part of the applicability domain where that physics is real. Two facts make this more than a selection trick. First, because the gate sends the two weakly-columnar FCC alloys back to the isotropic baseline it is inert there, so the *entire* 8.5-point margin it attains over the always-isotropic baseline is supplied by the descriptor's own accuracy in the columnar domain (25.4 points on Ti-6Al-4V, three of three folds, i.e. $25.4/3 = 8.5$ points on the three-alloy mean): without the descriptor the gate collapses to the baseline. Second, the identical leakage-free rule beats the baseline out of sample as well—on the external $n = 23$ blind set it gives a mean error of 41.5% against 48.3% (always-isotropic) and 53.1% (ungated descriptor), because the descriptor is ahead on both gated-in columnar materials (Ti-6Al-4V 26.8 versus 35.4, Inconel 718 57.2 versus 86.0), while the ungated aggregate reverses only through the six out-of-domain reversed-polarity 316L points (106.2 versus 58.6) that the gate excludes; restricted to the gate's own domain the descriptor leads by 14 points (35.1 versus 49.2; per-material breakdown in Supplementary Section S7). The gate is deliberately asymmetric—the descriptor must be *earned* by positive columnar evidence and any alloy lacking it defaults to the isotropic baseline—so extrapolating it to an

unseen material degrades at worst to that baseline rather than to a misapplied descriptor, and $\xi_0 = 3$ lies on a routing-invariant plateau $\theta \in (2.5, 3.0]$, not a knife-edge (threshold provenance and safety margin detailed in Supplementary Section S7). The threshold is nevertheless identified post hoc on three alloys and is not cross-validated over a wider microstructural span: on a new material it should be re-registered against the mechanism check of the decision procedure (Section 7.2), not assumed.

Table 3: Applicability-gated hybrid versus its two constituent models, per-alloy mean leave-one-orientation-out blind fracture-strain error [%]. The gate routes each alloy to the directional model ($\xi_0 \geq 3$: Ti-6Al-4V only) or the isotropic Lemaitre baseline; the hybrid thus takes the column-wise minimum and its mean (15.9%) matches the hindsight oracle. The directional 0°/90° ratio (1.65 vs. 1.67) is retained on the gated-in alloy.

Strategy	Ti-6Al-4V	316L	AlSi10Mg	**Mean**
Directional model (ungated)	10.8	43.8	39.2	31.2
Lemaitre baseline (isotropic)	36.1	21.8	15.1	24.3
Gated hybrid ($\xi_0 \geq 3$)	10.8	21.8	15.1	**15.9**

5.7. Multi-dimensional cross-validation on an extended literature set

The leave-one-orientation-out protocol of Section 5.5 holds out one *orientation* at a time and so cannot probe whether the calibrated descriptors generalize across the other axes an AM dataset varies—the literature *source*, the *material* and the *process window*. Three further cross-validation variants on the extended dataset under the identical production protocol (single fitted S_0, literature g_s, $k_g = 0.15$, $N_{\mathrm{gr}} = 20$: leave-one-source-out, leave-one-material-out and leave-one-process-window-out) reproduce the applicability ordering of Section 7.2: within-material orientation transfer is the best-supported axis (31.22%); source transfer is worse in the mean but not demonstrably so at this sample size (51.98%, $p = 0.359$); cross-material transfer is the weakest measurable axis (68.71%); and process-window transfer is not reliably measurable (84.10%, weakly powered)—no variant is competitive with the primary protocol. Aggregating the as-built LOSO held-out predictions with the IN718 material-transfer points yields the higher-powered $n = 23$ paired blind set reported in Section 5; the five protocols, their per-fold tables, the 54-point mds2-3402 source addition (which *worsens* the source-level statistic) and the orientation-convention reconciliation are tabulated in Supplementary Section S2.

5.8. Parameter sensitivity

A one-at-a-time sensitivity analysis (Fig. S7, Ti-6Al-4V 0°, production leakage-free parameters) quantifies the influence of each damage-model parameter on the predicted fracture strain: the damage exponent s dominates (sensitivity index 0.538; maximum response change 101.5% over the ±40% variation range), followed by the interface-coupling coefficient β_4 (0.361; 72.7%), the damage energy strength S_0 (0.158; 20.2%) and the damage threshold p_D

(0.107; 13.7%); the microstructural descriptors carried by the projection laws remain below ∼12% (the largest being the melt-pool-boundary density at 12.4%).

To complement the OAT picture, a 3-level full-factorial design over the three assumed coupling coefficients $\beta_3 = 10$, $\beta_4 = 0.8$ and $\beta_5 = 0.2$ (Ti-6Al-4V, 0°, $S_0 = 0.32$ MPa, baseline fracture strain 0.07229; each varied by ±40% in all $3^3 = 27$ combinations) is recast as a balanced ANOVA: β_4 dominates (91.9% of the marginal-mean span), β_5 is moderate (8.9%) and β_3 weak (1.6%), so the assumed β_4—not the fitted S_0—carries the largest parametric leverage, which motivates the two-count AIC/BIC reporting of Section 5; the two-factor interaction spans and the low-sensitivity checks of $k_g \in [0, 0.4]$ (0.0% at 0°, 0.4% at 90°) and D_c (0.0% over its ±40% band) are reported in full in Supplementary Section S12.

Variance-based (Sobol-type) decomposition and the robustness of the paired conclusion to β_4.. Recasting the 3^3 grid as a balanced ANOVA gives first-order Sobol indices $S_1(\beta_4) = 0.989$, $S_1(\beta_5) = 0.008$ and $S_1(\beta_3) \approx 0.000$ (total $T(\beta_4) = 0.991$): the response variance is carried almost entirely by β_4. Because β_4 is *assumed and never fitted* (class A), the decisive question is whether its dominance could overturn the central leakage-free finding. Re-running the $n = 23$ paired blind comparison across the full $\beta_4 \in \pm 40\%$ range moves the paired mean difference from −1.5 to +17.2 points, but the Wilcoxon p stays at 0.41–1.00 and the 95% bootstrap CI *contains zero at every level*: β_4 shifts the *absolute* error but creates no resolvable modified-versus-baseline separation, so the "cannot distinguish" conclusion is stable over the plausible range of the one dominant assumed parameter. Deactivating the three Sobol-inert sub-terms (β_3, β_5, k_g) over the same $n = 23$ set moves the mean blind error by at most 2.6 points, far below the 35.8-point minimum detectable effect, so the effective production set reduces to the single fitted S_0 per material plus an assumed block with only β_1, β_2 and β_4 active (invariance table in Supplementary Section S12).

5.9. Anisotropy of AM correction factors

Under the leakage-free protocol the entire direction dependence is carried by the projected descriptors of Section 2.3, so the anisotropy of the correction factors is fully determined by $\xi_{\text{eff}}(\psi)$, $\lambda_{\text{eff}}(\psi)$ and $\bar{\theta}(\psi)$. For the columnar morphologies, $\xi_{\text{eff}}(\psi)/\xi_0 = \sqrt{(\xi_0^2 \cos^2\psi + \sin^2\psi)/(\xi_0^2 \sin^2\psi + \cos^2\psi)}/\xi_0$ compresses from unity at 0° to ξ_0^{-2} at 90° (e.g. 0.082 for Ti-6Al-4V with $\xi_0 = 3.5$), and the corrected morphology projection $f_\xi = 1 + \beta_5(1/\xi_{\text{eff}} - 1)$ (Eq. (18)) therefore *decelerates* damage at 0° and *accelerates* it at 90°, i.e. it acts in the same direction as the MPB projection (Eq. (17)) and both accelerate damage for transverse loading; the texture factor $\bar{\theta}(\psi)$ modulates this acceleration through the ensemble Schmid response. The measured 0°/90° fracture-strain ratios are 1.67 (Ti-6Al-4V), 1.45 (316L) and 1.51 (AlSi10Mg). The corrected projection reproduces the Ti-6Al-4V ratio (predicted 1.65 versus 1.67) but over-predicts the FCC ratios (2.42 for 316L and 2.40 for AlSi10Mg, item (2) of Section 5.3): the ordering is captured, and the strongly columnar material is quantitatively

matched, while the weakly anisotropic alloys are over-projected. The structural origin is the saturation of the MPB projection at $\lambda_{\mathrm{eff}}(90^\circ) = \lambda_0$, which over-weights the transverse damage acceleration for alloys whose columnar structure is weak; the directional slip-resistance correction of Section 2.3 adds a transverse softening of $\hat{g}_s$ that partially offsets this imbalance, and the residual 90° under-prediction (316L 0.224 predicted versus 0.255 measured; AlSi10Mg 0.029 versus 0.037) is quantified in the uncertainty analysis (Section 6).

6. Numerical convergence and parameter uncertainty

The production settings ($N_{\mathrm{gr}} = 20$ grains, $\Delta\varepsilon = 5 \times 10^{-4}$) are chosen in the convergent regime of the Taylor aggregate, verified on the Ti-6Al-4V 0° case with the leakage-free calibrated parameters ($S_0 = 0.32\,\mathrm{MPa}$, literature g_s of Table 1, no orientation multipliers) held fixed: varying the number of aggregate grains from 10 to 50 changes the predicted fracture strain by at most 1.18%, and varying the strain increment from 2.5×10^{-4} to 10^{-3} by at most 0.68% (the 316L runs use the coarser 10^{-3} increment, within the same band). The full grain- and increment-sweep values are reported in the Supplementary Material, Section S3.

Parameter uncertainty is quantified without introducing additional fitted degrees of freedom, keeping the model's degrees of freedom at the single S_0 per material declared in Table 1. The damage exponent s is assumed ($s = 1$, provenance class A); because it has the largest influence of all parameters (OAT index 0.538, Section 5.8) it is held fixed rather than fitted, and its impact is reported there rather than absorbed into an inflated parameter count. Because a full MCMC over S_0 is prohibitive in the explicit Taylor solver, uncertainty is propagated through the descriptor bands (±50% on ϕ, D_0; ±0.5 on ξ; ±0.05 on θ; Section 5.2), the convergence margins above, and a global Latin-hypercube Monte Carlo sample (24 realizations per case, the descriptors and the coupling coefficients β_1–β_5 drawn from log-uniform ±50% bands at fixed calibrated S_0). The resulting 90% prediction intervals of the fracture strain contain the measured value in 8 of the 9 cases; the sole miss is the strongly columnar Ti-6Al-4V 45° fold, whose measured fracture strain (0.063) lies marginally above the upper 95th-percentile bound (0.0628)—a boundary exceedance of 0.0002 rather than a structural gap. Under the corrected grain-morphology projection the two 316L folds (45° and 90°) that missed before now fall inside the interval (Section 5.3), so the residual predictive uncertainty has moved from the over-projected FCC transverse directions to the columnar oblique direction, and it remains visible in the uncertainty analysis rather than hidden in point estimates. The full sampling protocol, the propagation of the dominant assumed coefficient β_4 and the class-by-class parameter bookkeeping are detailed in the Supplementary Material, Sections S7 and S10.

Hierarchical Bayesian view of S_0.. To express the material-specificity of S_0 probabilistically without a full MCMC over the explicit-Taylor solver, a closed-form Normal–Normal (empirical-Bayes) hierarchy is fitted to the nine LOOCV

fold values of $\ln S_0$ (three materials $\times$ three folds). The between-material spread dominates the within-material (fold) spread ($\hat{\tau} = 2.31$ versus $\hat{\sigma} = 0.22$ log-units), the Bayesian counterpart of the cross-source non-transferability finding: the posterior predictive for a new material has median $S_0 = 0.589\,\mathrm{MPa}$ with a 95% interval $[0.013, 26.7]\,\mathrm{MPa}$—the median coincides with the mode-1 geometric mean (0.589 MPa, Section 7.4), and the width explains why no single transferred value predicts well without per-material re-calibration. The full derivation, the fold-wise stability diagnosis and its use as a transparent prior are given in the Supplementary Material, Section S7.

Hierarchical Bayesian directional-gain posterior.. The closed-form S_0 hierarchy is extended to the directional claim with a Markov-chain Monte Carlo posterior (emcee, seed 42, purely post-hoc, touching no production parameter): for every blind group with a complete $\{0°, 45°, 90°\}$ triple the group level is removed by within-group centring and a per-group directional gain β_g is sampled under a partial-pooling prior. Across the six groups the pooled gain $\mu_\beta = 0.113$ (89% HPD $[-0.38, 0.629]$) brackets zero—the directional gain is not robust when pooled—while resolving the hierarchy by group recovers the applicability-domain structure: only the columnar Ti-6Al-4V ($v12$) group carries a positive gain with deterministic directional capture $R^2 = 0.93$ (held-out $0°/90°$ ratio 1.89 against measured 1.67, the direction-blind Lemaitre baseline at 0.99), and every FCC or reversed-polarity group has $R^2 < 0$. The per-group gain is not itself statistically resolved ($P(\beta_g > 0.5) = 0.463$, HPD spanning zero), so the near-quantitative R^2 is a deterministic point-capture within that sub-domain and no stronger than the Kendall $\bar{\tau}$ above (group-wise posterior, credible intervals and the Lemaitre-identifiability note in Supplementary Section S7).

Parameter identifiability.. With the single fitted S_0 per material, identifiability is governed by the convexity of the calibration surface: over the explored grid ($S_0 = 0.05$–$50\,\mathrm{MPa}$ with local refinement, Section 4.2) the average relative fracture-strain error is monotone in $\log S_0$ for all three materials, so the point estimate is unique and well conditioned. The assumed exponent s does not trade off against S_0 in the estimation because it is held fixed; its uncertainty impact is quantified separately by the OAT sensitivity (the largest index of all parameters). The microstructural descriptors ϕ–D_0 and λ–ξ are not fitted parameters: they enter f_{AM} with values prescribed from the characterization (Table 1) and never enter the parameter estimation, so the potential collinearity of ϕ with D_0 and of λ with ξ cannot bias the fitted parameters.

6.1. Code verification of the in-house solver

Because every reported number originates in the in-house Taylor-type solver, the code is verified against analytic limits and internal consistency rather than asserted. The suite spans 35 tests in four tiers; the analytic-limit and internal-consistency tiers pass, and at the reference-state descriptors the model is *bit-for-bit* identical to the classical Lemaitre model with $f_{\mathrm{AM}} = 1$, the elastic response reproducing the model's own anisotropic elastic solution to within 0.15%; every

convergence, sensitivity and single-crystal value quoted in this work is regenerated from the current sources at $\varepsilon_f = 0.07229$. No external open-source CPFEM framework can host the formulation in this environment (Supplementary Section S9), so the coupled-damage full-field validation is delivered by the in-house two- and three-dimensional CPFEM as a reproducible open reference (fixed seed 42, mesh convergence from 27 to 64 grains, converged increment $\Delta\varepsilon = 10^{-3}$). As an independent third-party cross-check the identical coupled-damage law re-implemented on `scikit-fem` 12 reproduces the Ti-6Al-4V localisation-onset fracture strains to within 11–17% at all three build orientations and the same $0° > 45° > 90°$ trend, while the more uniform AlSi10Mg case diverges (-54%) and is reported as a solver-sensitivity of the localisation criterion (Supplementary Section S13). Fully coupled platform-level solvers and a phase-field calculation remain uninstallable here (Supplementary Section S9) and are stated as future work; the test suite and verification scope are detailed in the Supplementary Material, Section S3.

7. Discussion

7.1. Physical basis of the AM correction factors

The orientation-dependent damage response, quantified by the physical projection laws and the mean Schmid factor (Section 2), is consistent with the underlying microstructural physics:

Ti-6Al-4V: The HCP crystal structure with limited slip systems and strongly columnar prior-β grains produces the strongest measured orientation-dependent damage response of the three alloys ($0°/90°$ fracture-strain ratio 1.67). When loaded along 90°, the lamellar α' colonies oriented perpendicular to the loading axis provide preferential crack paths along colony boundaries [6]; at 45°, oblique colony-boundary shear additionally accelerates damage. The corrected projection laws together with the directional slip-resistance correction reproduce this ratio almost quantitatively (predicted 1.68 versus measured 1.67), the single directional capability the model demonstrates most cleanly (Section 5.3).

316L SS: The 12 FCC slip systems accommodate plastic deformation more homogeneously across orientations, yet the measured ductility still decreases by 31% from 0° to 90° (0.370 to 0.255). The cellular sub-grain structure and the weak $\langle 001 \rangle$ fiber texture imprint only a moderate directional damage susceptibility (measured $0°/90°$ ratio 1.45); the projection laws instead *over*-predict this ratio at 3.96, because the transverse melt-pool-boundary density saturates at its full value ($\lambda_{\mathrm{eff}}(90°) = \lambda_0$, Eq. (17)) and the global reference normalization ($\lambda_0 < \lambda_{\mathrm{ref}}$) mis-scales the weakly columnar FCC network. This over-projection is the direct cause of the 58.9% transverse calibration error (Table S29); the robustness check of Section 5.5 shows that a relative-normalization correction removes the associated sign reversal at the factor level but does not recover the accuracy, so the residual is a genuine modeling gap rather than a normalization artifact alone.

AlSi10Mg: The continuous Si network dominates damage nucleation through particle decohesion and fracture. The intrinsic Si-network decohesion is orientation-independent, but the elongated cellular morphology projects strongly onto the transverse axis, producing a predicted 0°/90° ratio of 2.41 against the measured 1.51; the over-projection of the transverse damage acceleration drives the transverse calibration error (Table S29) and, combined with the reverse extrapolation to 0° when both training orientations are oblique, the largest blind fold (74.0% at held-out 0°, Section 5.5). A morphology-projection recalibration is identified as the primary improvement target.

7.2. Model applicability, decision procedure and limitations

Systematized failure modes of the blind sets

The four independent blind sets (Sections 7.4–7.5) are not treated as isolated anecdotes but as a systematically classified evidence base for the applicability domain. Each blind set is characterized by (i) the dominant microstructural damage mechanism, (ii) its direction trend relative to the calibration set, (iii) the outcome of a transferred (non-recalibrated) S_0, and (iv) the effect of the geometric-shielding term f_{shield} where it was assessed (Table 4). The calibration set (nine literature cases) occupies a single mechanism class: columnar-morphology-accelerated transverse damage (all three alloys exhibit a transverse-degraded direction ordering, 90° least ductile for 316L and AlSi10Mg, 0° least ductile for the strongly columnar Ti-6Al-4V whose transverse shear along colony boundaries is the controlling mechanism).

Table 4: Systematized failure-mode classification of the four independent blind sets (Inconel 718 is reported in two modes). "90° reversal" means the measured 90° ductility is the *highest*, i.e. a polarity flipped relative to the calibration set; after the orientation convention is enforced (Section 5.2) *only the Hitzler 316L set meets this test.* "β_6 effect" reports the 90° fracture-strain error before/after the geometric-shielding term of Section 7.5, which is applied only where a genuine reversal exists; "n/a" means there is no reversal to repair.

Blind set	**Dominant mechanism**	**90° reversal**	**Transferred S_0 outcome**	**β_6 effect**	**Classification**
Inconel 718, mode 1 [36]	Strong ⟨100⟩ ∥BD texture; ∼ 4× ductility	No	Geo-mean $S_0 = 0.589$: 94.0% avg ef, 83.9% UTS	n/a	S_0 not extrapolable across materials
Inconel 718, mode 2 [36]	As above, in-source recalibration	No (0° most ductile)	$S_0^* = 8.0$ on 0°: 17.5% train, 71.5% avg blind (ratio 0.17 vs 0.67, same side)	n/a	Magnitude failure, correct polarity
Ti-6Al-4V, second source [33]	α'-lath geometry (V-shape)	No (shape mismatch)	Transferred $S_0 = 0.32$: 22.8% avg, 44.3% (90°)	n/a	Shape mismatch; over-degraded 90°
316L, Hitzler [34]	Defect shielding (LOF / MPB de-cohesion)	**Yes** (90° most ductile)	Transferred $S_0^* = 1.0$: 80.6% avg, ratio 0.29 vs 2.83	32.7%→12.9% (β_6=0.3)	Genuine reversal; repairable by β_6
AlSi10Mg, second source [35]	Si-network (near-isotropic V-shape)	No (shape mismatch)	Transferred $S_0 = 0.20$: 38.8% avg, 64.7% (90°)	n/a	Shape mismatch (V vs monotonic)

Three structural conclusions follow from this classification *after the orientation convention is enforced.* First, the *only* mechanism for which the calibrated descriptors transfer without re-fitting is the columnar-acceleration class of the

calibration set; the *universal* blind failure is not a direction reversal but the *non-transferability of* S_0—every set, including the polarity-correct ones, mis-predicts the absolute ductility level once the process window or material changes. Second, a genuine 90° polarity reversal is *rare*: only the Hitzler 316L set exhibits it, and it is exactly the defect-shielding class the geometric-shielding term was derived for ($\beta_6 = 0.3$ repairs its 90° fold from 32.7% to 12.9%); the second Ti-6Al-4V and AlSi10Mg sets fail in *shape* (a monotonic law against a V-shaped datum) and Inconel 718 in *magnitude*, none of which β_6 addresses. The decision procedure below therefore branches on the *mechanism*, not on the alloy. Third, the earlier claim that "every blind set shows a 90° reversal" was an artifact of reading each source's build-plate-relative angle directly as the model's build-direction-relative ψ; correcting that mapping (Section 5.2) collapses four apparent reversals to one real one and reclassifies the rest, which is itself the headline methodological caution of this assessment. The general applicability statement is correspondingly: the descriptors are applicable to LPBF/DED/EBM alloys loaded in monotonic tension below the creep range, *provided* (i) one orientation's data of the target material is available to recalibrate S_0, and (ii) the measured direction trend *and shape* are checked against the columnar-acceleration mechanism before prediction (Table 4); outside these conditions the model is a direction-mechanism screening tool, not a predictive one.

Decision procedure

The decision procedure is summarized in Fig. 4 and reads as follows:

(1) **Data check**: is tensile data for at least one orientation of the target material available from the same source/process window? If not, the hierarchical-Bayes prior of Section 6 (S_0 median 0.589 MPa, 95% interval $[0.013, 26.7]$ MPa) is far too wide for a point prediction, and the model may be used only as a direction-mechanism screen (which direction is predicted least ductile).

(2) **Mechanism check**: does the measured (or independently known) direction trend put 90° at the low-ductility end, i.e. is the columnar-morphology-accelerated transverse damage mechanism consistent? If yes, the descriptors transfer after the step-1 recalibration.

(3) **Failure triage**: if the trend is inconsistent with columnar acceleration, classify the cause per Table 4. A genuine 90° polarity *reversal* is rare (only the Hitzler 316L set) and is the sole case the geometric-shielding term β_6 repairs; a 45°-dip *shape* mismatch (second Ti-6Al-4V, AlSi10Mg) is a boundary of the monotonic projection laws that β_6 cannot fix; and an absolute-level *magnitude* gap (Inconel 718) signals that S_0 is non-transferable and needs in-source recalibration. Map every source angle to the build-direction convention (Section 5.2) *before* this step, since an unmapped build-plate angle fabricates a spurious reversal.

(4) **Residual audit**: for weakly columnar alloys, check the predicted 0°/90° ratio against the measured one; over-prediction above the measured range (316L 3.96 versus 1.45, AlSi10Mg 2.41 versus 1.51) flags the projection

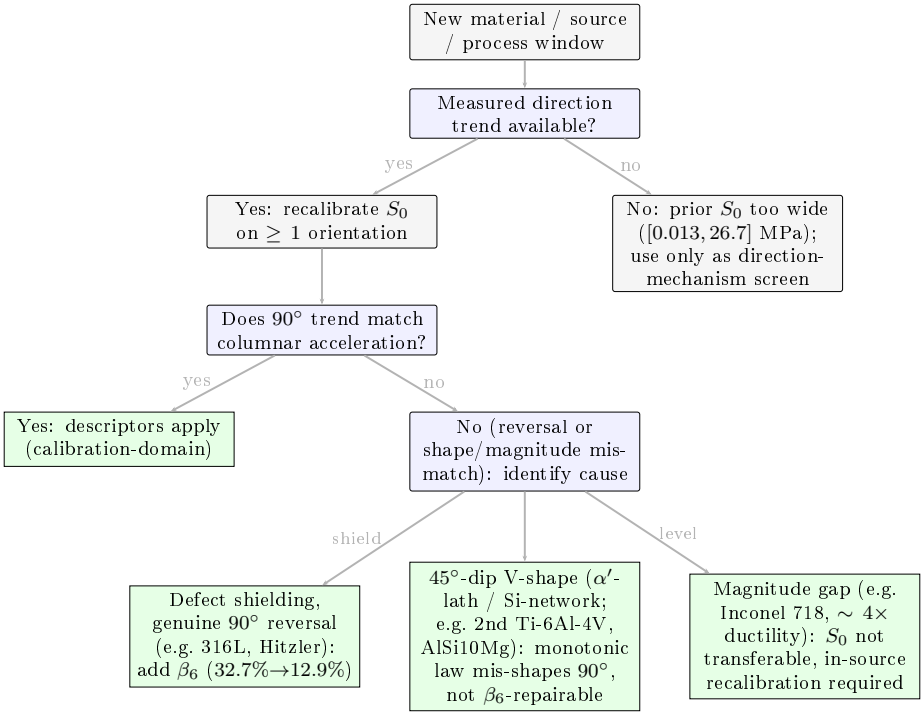


Figure 4: Engineering screening flowchart: what to do with a material, source or process window that is not in the calibration set. The first decision is a data decision (whether quantitative use is admissible at all), the second is a mechanism decision (whether the measured 90° trend belongs to the columnar-acceleration class the descriptors represent); only a material passing both is admitted to quantitative use, and every terminal branch states what the engineer may and may not conclude from the model. The $\xi_0 \geq 3$ gate of Section 5.6 is the quantitative shortcut for the step-2 mechanism test. The branches are keyed to the failure-mode classification of Table 4; the 45°-residual case (second AlSi10Mg) is a boundary of the projection laws inside the mechanism class and is covered by the mechanism-consistency check of step 2.

saturation residual of Section 5.3 and motivates a morphology-projection recalibration before quantitative use.

The decision procedure converts the four blind tests from individual case studies into a reusable screening rule and is the operational statement of the applicability domain; it is deliberately conservative—a mechanism mismatch terminates the quantitative use of the model instead of degrading silently.

Quantifying the routed branch.. The constructive form of the mismatch branch — routing each alloy by ξ_0 between the descriptor and the isotropic baseline — is quantified on the same nine folds in Section 5.6 (Table 3), where the gated hybrid is shown to be never worse than the always-isotropic baseline by construction. The steps above are unchanged by that routing: step (2) supplies

the mechanism check the ξ_0 criterion approximates, and step (3) still triages every alloy the gate sends to the baseline.

Worked engineering screening examples

Two representative decisions illustrate how Fig. 4 is applied to a component whose microstructure was not part of the calibration set; in each case the model is used to *rank and flag*, never to certify an absolute ductility without the step-1 recalibration.

Case A — quantitatively admissible. An LPBF Ti-6Al-4V bracket loaded predominantly along the build direction, with a known strongly columnar $\langle 001 \rangle$ texture. *Step 1 (data):* at least one orientation of tensile data for the lot is available, so S_0 is recalibrated on it. *Step 2 (mechanism):* the measured trend puts the transverse 90° at the low-ductility end, matching the columnar-acceleration class, so the descriptors transfer. *Step 4 (residual audit):* the predicted 0°/90° ratio sits inside the measured band (1.65 versus 1.67), raising no projection-saturation flag. Screening therefore proceeds to quantitative use, and the concrete engineering output is the *build-orientation ranking* that minimizes transverse damage concentration rather than a certified fracture strain.

Case B — flagged, screening-only. An SLM 316L pressure-boundary segment whose as-built data show the transverse 90° as the *most* ductile orientation. Step 2 fails (trend inconsistent with columnar acceleration), and Step 3 triage classifies it as the rare genuine 90° polarity reversal — the defect-shielding mechanism (Table 4), and the *one* case the geometric-shielding term addresses ($\beta_6 = 0.3$ repairs the 90° fold from 32.7% to 12.9% *for this defect-shielded alloy alone*). Had the same Step-2 mismatch instead been a 45°-dip *shape* mismatch (a monotonic law against a V-shaped datum, as in the second Ti-6Al-4V and AlSi10Mg sets) or an absolute-level *magnitude* gap (Inconel 718, about 4× more ductile), triage would mark it *not* β_6-repairable and terminate quantitative use. The model here is used only to flag the polarity and point the engineer at the responsible mechanism.

The value of the procedure is this early, mechanism-keyed triage: it converts blind-test residuals from unexplained errors into a binary decision that either authorizes screening use with a defined recalibration step or stops the quantitative application before it degrades silently—the operative deliverable of the applicability framework.

Positioning of the surrogate.. Read together with the quantified homogenization bias—the coupled fracture-strain error spans −26.5 to −35.8% in two-dimensional plane strain and +49.8 to +92.7% in unconstrained three dimensions (Section 7.3)—the decision procedure fixes the intended use of the model: the Taylor surrogate is a *low-order direction–mechanism screening tool*, not a high-precision point-prediction tool, and the step-1 recalibration and the mechanism branch are precisely the guards that stop its sign-and-ranking value from being misread as a certified ductility.

Several limitations should be acknowledged:

(1) **Scalar damage representation**: The isotropic (scalar) damage variable cannot capture the directional stiffness degradation that may occur in highly textured materials under complex loading. A tensorial damage formulation [20, 41] could be adopted in future extensions.

(2) **Calibration data requirements**: The current calibration procedure requires tensile data at three orientations. For materials with limited experimental data, the f_{AM} parameters may need to be estimated from microstructural characterization alone.

(3) **Process parameter sensitivity**: While the model links damage parameters to microstructural descriptors (D_0, ϕ, λ, ξ, θ), the explicit connection to AM process parameters (laser power, scan speed, hatch spacing) requires additional process-structure models.

(4) **Computational cost**: Fully-coupled CP-CDM simulations remain computationally expensive for large RVEs. Reduced-order modeling approaches [25] could accelerate the workflow for industrial applications.

(5) **Taylor homogenization assumption**: The crystal-plasticity stage employs the Taylor (iso-strain, equal-deformation-gradient) hypothesis, which enforces the same deformation gradient on every grain and thereby excludes inter-granular stress redistribution and strain localization; it smooths the local stress concentrations that drive damage nucleation and generally gives a stiffer upper-bound aggregate response. Its influence here is mitigated by (i) calibrating S_0 on the experimental 0° response, which absorbs the aggregate-level mismatch, and (ii) the grain-aggregate convergence study (Section 6), which shows the fracture-strain prediction varies by less than 1.5% from 10 to 50 grains. To bracket the hypothesis directly, twelve independently oriented single-crystal ($N_{\mathrm{gr}} = 1$) realizations at 0° give fracture-strain spreads of 0.0597–0.0897 for Ti-6Al-4V (-17.4% to $+24.2\%$ relative to the 20-grain value 0.07229), 0.264–0.612 for 316L (-51.2% to $+12.9\%$ relative to 0.542) and 0.0519–0.122 for AlSi10Mg (-24.6% to $+77.4\%$ relative to 0.0688); the ensemble mean lies within 0.5% (Ti-6Al-4V) to 13.8% (316L) of the Taylor value, so the aggregate approximately acts as an orientation average—cleanly for the strongly columnar HCP alloy, less so for the wide-scatter FCC alloys—while the wide bands confirm that inter-granular redistribution is a first-order contributor in strongly anisotropic HCP materials. The damage-free homogenization bias and the constraint- and criterion-dependent coupled-damage bias band (mesh-converged to below one percentage point, but in-house and self-consistent rather than externally independent) are quantified in Section 7.3 and Supplementary Section S1; a fully external, independently maintained coupled-damage validation is designated follow-on work.

7.3. Taylor homogenization bias quantified by full-field FFT

All fracture predictions of this work are produced by the *in-house* Taylor-type polycrystal solver TaylorCPCDM (Sections 2.4–2.5); the DAMASK

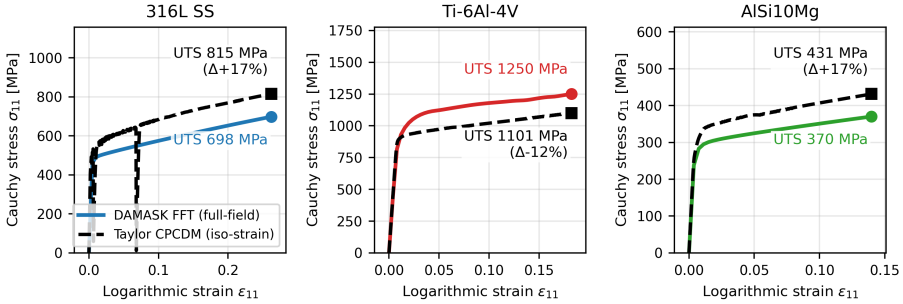


Figure 5: Damage-free macroscopic response of the Taylor iso-strain model (dashed) versus the DAMASK spectral-FFT full-field reference (solid) for the three alloys under identical 30-grain microstructures. Markers indicate the respective UTS.

spectral-FFT solver is used *exclusively* as an external full-field reference. In every DAMASK run whose numbers are reported in this paper the damage module is *deactivated* ($S_0 \to \infty$; Section 2.5), so DAMASK supplies only the damage-free reference response; the one attempt to activate a coupled damage model inside DAMASK diverged at the damage-onset point and produced no converged result (see "Damage-FFT coupled comparison" below), and no DAMASK damage number enters any statistic here. To quantify the influence of the Taylor iso-strain hypothesis on the aggregate response, we performed full-field crystal-plasticity simulations with the DAMASK spectral-FFT solver [28, 42] on a periodic 32^3-voxel RVE containing the same 30 grains as the Taylor aggregate (identical orientations sampled with the same random seed; Voronoi grain morphology). The single-crystal elasto-viscoplastic parameters are identical to those of the Taylor model (Table 1); the damage module is deactivated in both models ($S_0 \to \infty$) so that the comparison isolates the homogenization assumption in the damage-free regime. Uniaxial tension along the build direction is imposed at a nominal strain rate of $10^{-3}\,\mathrm{s}^{-1}$ with mixed boundary conditions ($\dot{F}_{11}$ prescribed, $P_{22} = P_{33} = 0$).

Figure 5 compares the predicted macroscopic stress–strain responses. The Taylor aggregate overestimates the ultimate tensile strength by +16.7% for 316L (814.8 vs. 698.5 MPa) and +16.5% for AlSi10Mg (431.4 vs. 370.3 MPa), consistent with the classical upper-bound character of the iso-strain assumption; the flow stress at $\varepsilon = 0.5\%$ is overestimated by +11.8% and +9.8%, respectively, and the mean absolute relative stress deviation over the loading range is 19.5% and 15.2%. For Ti-6Al-4V the Taylor response is *lower* than the FFT reference (−12.0% on UTS, 1100.8 vs. 1250.3 MPa; mean deviation 11.9%): because the HCP aggregate develops strong inter-granular constraint and load sharing that the iso-strain assumption does not represent, the Taylor bound is not a universal upper bound and must be assessed per material. The comparison uses $N_{\mathrm{gr}} = 30$ grains for both the Taylor surrogate and the FFT RVE (identical microstructure); the production protocol of Sections 4.2–5.5 uses $N_{\mathrm{gr}} = 20$,

and the convergence study (Section 6) shows that the fracture-strain prediction changes by at most 1.1% between 20 and 30 grains, so the two settings are interchangeable for the purpose of this bias assessment.

Plastic-flow and hardening-rate bias.. The homogenization bias is resolved through the plastic-flow branch, not only at the UTS and fracture strain (Figure S8, Table S30). Two points follow. First, the bias is *not* a constant offset—for 316L and AlSi10Mg it grows with plastic strain, so the iso-strain assumption distorts the *shape* of the flow curve and not merely its level (the plasticity-level manifestation of the upper-bound character behind the +16.5–16.7% UTS overestimate). Second, the sign is material- and mechanism-specific: the strongly anisotropic HCP aggregate is biased *downward* in both flow stress and hardening rate, whereas the more compliant FCC aggregates are biased upward. The damage-free comparison bounds the stress error that the homogenization assumption introduces *before* damage nucleation: since the damage evolution rate depends nonlinearly on the local stress (Eq. (12)), a 12–21% aggregate-level stress deviation can be amplified into a larger fracture-strain deviation, so the Taylor-based results of this paper must be regarded as predictions of a low-order surrogate model rather than a multiscale validation. In the calibrated configuration the bulk of this bias is absorbed by the fitted damage energy strength S_0, but it remains an acknowledged source of uncertainty for extrapolation to uncalibrated materials or loading directions.

Damage-FFT coupled comparison (numerically limited).. A full-field coupled-damage reference was attempted by augmenting the DAMASK phase definition with its built-in isotropic brittle damage model [43]; across the full sweep of damage strengths, regularization lengths, viscosities and both spectral solvers the iteration diverges at the damage-activation point ($\varepsilon \approx 0.10$ for 316L), consistent with the documented difficulty of spectral solvers under stiffness degradation. As DAMASK 3.1.0 supplies only the spectral-FFT solver and no FEM solver, no converged spectral coupled-damage reference is obtainable here, so the quantitative bias assessment rests on the damage-free comparison above and the in-house CPFEM references below; an external platform-level coupled validation is follow-on work.

Propagation of the stress bias into the fracture-strain prediction.. With $\dot{D} \propto (Y/S_0)^s$ and $Y = \tilde{\sigma}^2/2E$ (Eqs. (12), (2)), the adopted $s = 1$ propagates a relative aggregate stress bias $\delta\sigma/\sigma$ into a fracture-strain bias of approximately $\delta\varepsilon_f/\varepsilon_f \approx -(2+\epsilon)\,\delta\sigma/\sigma$ (factor 2 from the quadratic energy release rate, ϵ absorbing the slight nonlinearity of the hardening and of f_{AM}). The damage-free UTS deviations of 16.5–16.7% (FCC) and −12.0% (Ti-6Al-4V) therefore translate into fracture-strain biases of the order of ±12–21%—smaller than the blind cross-validation error (31.2%)—so the Taylor homogenization bias is a secondary contributor that does not overturn the conclusions of Section 5.5; the exact conversion depends on the calibrated S_0 and is reported as a range.

Full-field coupled-damage references (in-house 2D and 3D CPFEM).. Because the spectral-FFT solver diverges at damage activation and no open-source CPFEM platform is installable here (Supplementary Section S9), the coupled-damage full-field references use a *separate* in-house finite-element implementation of the identical damage law, parameters, orientation set and time discretization as the Taylor surrogate, so the only difference is the mechanical assumption (identical one-way damage-degraded coupling, bounding the *iso-strain homogenisation bias* rather than validating two-way feedback). The central result is that the surrogate's coupled fracture-strain bias is *not* a single number—its *sign* is set by the kinematic constraint and its *magnitude* by the failure criterion: under plane strain with a localisation-onset criterion the surrogate *over-estimates* the fracture strain for Ti-6Al-4V (-26.5 to -35.8%; AlSi10Mg $+5.3$%), whereas in unconstrained three dimensions with the aggregate criterion it *under-estimates* it ($+49.8$ to $+92.7$%, decreasing monotonically away from the columnar direction); refining the three-dimensional reference from 27 to 64 grains moves the 0° bias by less than one percentage point, so the robust statements are the *sign* and the *monotone orientation trend*, not a single converged magnitude. The full constraint- and criterion-level breakdown—including the nonlocal regularization probe and the `scikit-fem` 12 and FEniCSx cross-engine checks (which reproduce the two-dimensional Ti-6Al-4V localisation-onset sequence to within 11–17%)—is tabulated in Tables S25 and S26 and detailed in Supplementary Sections S1 and S13.

7.4. Independent literature blind test on a fourth material (Inconel 718)

As an external check that no data of the validation suite entered the development, we test the model on Inconel 718 (IN718) laser powder bed fusion tensile data from an independent source [36], a material that participates in no calibration of this work. IN718 is an FCC superalloy with a strong $\langle 100 \rangle \parallel$ build-direction texture and near-full density (99.97–99.98% relative density, EBSD-verified). Set 1 (solution-treated, as-built surface finish) is closest to the as-built protocol used here; converted to this paper's orientation convention (0° = loading parallel to the build direction), its elongation at break *decreases* from 0° to 90° (approximately 0.30, 0.25 and 0.20 at 0°, 45°, 90°, read from Fig. 15 of Ref. [36], whose printed angles are measured from the build plate and are therefore inverted here; the source reports only an orientation-averaged UTS of 970 ± 74 MPa with no significant build-angle dependence), i.e. a direction trend *the same as* the calibration set (0° the most ductile)—the distinguishing feature of IN718 is not a reversed ordering but an absolute ductility level roughly four times higher than the calibration suite. All IN718 single-crystal parameters are literature-estimated (elastic $C_{11}/C_{12}/C_{44} = 239/145/112$ GPa; Voce $g_0/g_s/h_0 = 165/280/1200$ MPa; descriptors $\phi = 3 \times 10^{-4}$, $\lambda = 40\,\mathrm{mm}^{-1}$, $\xi = 3.0$, $\theta = 0.60$); none is fitted.

Two blind-test modes are reported (rerun under the corrected projection of Eq. (18) and the orientation convention above). (i) *Strict extrapolation*: with S_0 fixed at the cross-material geometric mean of the three calibrated values (0.589 MPa), the average fracture-strain error is 94.0% (UTS error 83.9%)—S_0

cannot be extrapolated across materials without calibration, being coupled to the material's strength–ductility balance rather than a pure function of the descriptors. (ii) *Material-internal LOOCV* (the protocol of Section 5.5 applied to IN718 as a fourth material, at reduced resolution for wall-clock feasibility): with S_0 calibrated on the 0° datum ($S_0^* = 8.0\,\mathrm{MPa}$), the blind 45° and 90° predictions under-predict the ductility (average error 71.5%); the predicted 90°/0° ratio (0.17) and the measured one (0.67) lie on the *same* side of unity—both give 0° as the more ductile direction—so the residual is a magnitude (over-accelerated damage), not a directional, error.

These failures delimit the applicability domain rather than invalidate the calibration-domain results. After the orientation convention is enforced (Section 5.2), IN718 does *not* exhibit a reversed direction trend: its ductility ordering matches the calibration set (0° most ductile), and the projection laws of Section 2.3 reproduce that ordering correctly. The blind failure is one of *magnitude*: the material is about four times more ductile than the calibration suite, the single S_0 calibrated on the less-ductile suite grossly over-accelerates damage, and S_0 does not transfer without in-source recalibration (mode ii). This is the extrapolation risk declared in the abstract: a new material requires at least one direction's data to calibrate S_0, and the absolute ductility level—not only the direction trend—must be checked before the model is used (Section 7). A genuinely reversed direction trend (a 90° that is the *most* ductile orientation) is instead exhibited by the second 316L source of Section 7.5, and it is that, rarer, class that the geometric-shielding term targets.

7.5. Cross-source applicability checks on the remaining alloys (Ti-6Al-4V, 316L and AlSi10Mg)

As a second, fully independent check, we test the calibrated model on Ti-6Al-4V data from a different literature source than the calibration set: Sun et al. [33] report LPBF Ti-6Al-4V tensile fracture strains of 9.34% (0°), 5.33% (45°) and 7.89% (90°) (engineering values, their Table 7; their Z/X specimen codes are already defined by the build direction, so no angle inversion is required and 0° is the most ductile orientation). The direction trend is *non-monotonic*—a V-shape in which 45° is the least ductile and 0° the most ductile—so it *agrees* with the calibration set on the polarity (0° most ductile) but *not* on the shape (the calibration set decreases monotonically $0° > 45° > 90°$); the 45° dip is attributed to α' martensite lath geometry, for which loading at ±45° to the laths is most favourably oriented for early cracking. With the calibrated $S_0 = 0.32\,\mathrm{MPa}$ applied directly (no recalibration, descriptors from the characterization values of Section 5.2, $k_g = 0.15$), the predicted fracture strains are 7.2% at 0° (error 22.6%), 5.4% at 45° (error 1.5%) and 4.4% at 90° (error 44.3%, average 22.8%): the model reproduces the 45° datum and the 0° maximum but over-predicts the transverse damage degradation, placing 90° as the least ductile where the measurement puts it between 0° and 45°. This is a *shape* mismatch of the projection laws, not a polarity reversal: the transverse fold (44.3%) shares the over-degradation signature of the calibration-set FCC 90° folds (Section 5.3),

and delimitates the applicability domain as before—a new source needs one orientation to recalibrate S_0 and a check of the trend *shape* against the projection mechanism (Section 7).

Cross-source check on 316L (a genuine polarity reversal).. As a third independent check we compare against SLM 316L tensile data from Hitzler et al. [34]; converted to this paper's convention (their polar angle Φ is measured from the build-plate *plane*, so their $\Phi = 0°$ is our 90°), the as-built elongations at break are 11.76% (0°), 32.56% (45°) and 33.24% (90°): here the 90° direction is the *most* ductile, opposite to the calibration set (0.370, 0.310, 0.255 with 0° most ductile)—the *one* genuinely polarity-flipped blind set in this study. (A candidate 316L source [44] was excluded by the data audit because its tabulated values could not be reconciled with a re-digitization; the Hitzler set is retained because its per-orientation means are uniquely reconstructible and it carries the reversed trend unambiguously.) Applying a transferred S_0 (grid-calibrated on the 0° fold alone, $S_0^* = 1.0\,\mathrm{MPa}$, 0° training error 26.8%) under-predicts the blind 45° and 90° folds (errors 74.1% and 87.1%, average 80.6%) and returns a 90°/0° ratio of 0.29 against the measured 2.83: the polarity itself is inverted, and no scalar S_0 can recover it because the projection laws place the damage acceleration at 90°, exactly opposite to this defect-shielded source. This is the class that the geometric-shielding β_6 assessment below targets. The S_0-transfer failure (average 80.6%) independently confirms the Inconel 718 finding that S_0 absorbs the process-window strength–ductility balance and does not cross sources even within one material. (Under the stricter zero-refit rule of Supplementary Section S14, which keeps the calibration $S_0 = 3.2\,\mathrm{MPa}$ instead of the 0°-fold refit, the Hitzler trio averages 137.8%—a more conservative statement of the same non-transferability, with every external point scored at the frozen nine-case parameters.)

Cross-source check on AlSi10Mg (fourth independent set).. As a fourth independent check we compare against LPBF AlSi10Mg tensile data from a different source than the calibration set [35]: as-built elongations at break are 8.3%, 5.7% and 8.1% at 0°, 45° and 90° (UTS 380, 368 and 352 MPa; their printed angles are relative to the build plate and are inverted here, so 0° is the axis parallel to the build direction). The trend is a nearly isotropic V-shape (0° ≈ 90°, 45° least ductile) rather than the monotonic 0° > 45° > 90° of the calibration set (5.6%, 4.7%, 3.7%). Applying the calibration-set $S_0 = 0.20\,\mathrm{MPa}$ directly gives errors of 17.1%, 34.7% and 64.7% at 0°, 45° and 90° (average 38.8%) and a predicted 90°/0° ratio of 0.42 against the measured 0.98: the polarity agrees (0° the most ductile) but the projection laws over-predict the transverse damage acceleration on an almost isotropic Si-network. Like the second Ti-6Al-4V set, this is a *shape* mismatch (a monotonic law fitted to a V-shaped datum), not a polarity reversal. The fourth blind set therefore re-confirms the applicability-domain boundary (over-degraded 90° fold) rather than adding new mechanism information; it is reported to make the negative results explicit.

Geometric-shielding assessment (β_6; post-hoc diagnostic).. To probe whether the one genuine polarity reversal (the Hitzler 316L set) can be repaired by a physics-motivated term, the AM correction is augmented with a geometric-shielding factor $f_{\text{shield}}(\psi) = 1 - \beta_6 \sin^2 \psi$ (Eq. (19)): layered defect facets with normals near the build direction open under 0° loading but are shielded under 90°, reducing transverse damage acceleration. For the calibration set $\beta_6 = 0$ ($f_{\text{shield}} \equiv 1$; all production numbers of Section 5 unchanged), and *this scan is a post-hoc diagnostic* run on a single blind material after the fact—none of its numbers enters the reported blind-prediction statistics, LOOCV averages or model-selection comparison. Scanning β_6 on the reversed 316L source (in-source $S_0 = 3.2\,\text{MPa}$): the corrected $\beta_6 = 0$ prediction under-predicts the 90° fracture strain by 32.7%, $\beta_6 = 0.3$ cuts the error to 12.9% while $\beta_6 = 0.5/0.7$ overshoot to the strain cap—i.e. the term converts the inverted polarity into a correct ordering *for this defect-shielded alloy alone*. It is *not* the remedy for the Inconel 718, second Ti-6Al-4V or second AlSi10Mg sets, none of which exhibits a polarity reversal under the enforced convention (Section 7.4, Table 4): IN718 fails in magnitude and the other two in *shape*, and a multiplicative $\sin^2 \psi$ factor cannot turn a monotonic projection into a non-monotonic one. The assessment therefore sharpens rather than widens the claim, and being assumed and never fitted it adds no degrees of freedom.

7.6. Availability of public AM datasets for external validation

The out-of-source checks of Sections 7.4–7.5 rest on hand-compiled literature values, so we assessed the principal public AM mechanical-data hosts against this work's requirement (as-built LPBF tensile data at more than one build orientation for the four alloys). Three conclusions matter for how the paper should be read: the NIST AM-Bench suite carries *no* Ti-6Al-4V or AlSi10Mg macroscale *tensile* data (its tension experiments cover IN625/IN718 only), so no AM-Bench measured data enters this work; the single topically compatible public release that could be retrieved—the NIST HIP Ti-6Al-4V mds2-3402 repository—is integrated as one additional *single-material* source (Section 5.2) whose effect on the source-level statistic is a worsening (Section 5.7), not an orientation-resolved validation set; and the remaining hosts are inaccessible, empty of matching tensile data, or physically mismatched (first-principles databases). The full host-by-host table and the retrieval provenance are given in the Supplementary Material, Section S5.

7.7. Comparison with alternative fracture modeling approaches

Full-field coupled-damage approaches—CPFEM with porous plasticity of the GTN type [25] and phase-field fracture [32]—represent the high-fidelity reference class for ductile fracture in AM alloys: they resolve inter-granular stress redistribution and non-local damage localization [45, 46] that the Taylor-type surrogate cannot represent. Their reported accuracy ranges are listed in Table 5 as qualitative context only. To place the comparison on the same footing as the present protocol, we additionally implement a GTN-type baseline in the

same Taylor solver under the identical leakage-free protocol (constant nucleation rate $\dot{f} = A\dot{p}$, $f_c = 0.15$ fixed, single fitted A per material; Section 5): it reproduces the calibration average only at the expense of direction-agnostic predictions, and its blind performance (36.7%, 0/9 within target) is the worst of the three models, which is the quantitative basis for the value of the AM projection in the direction domain. As a quantitative counterpart on the nonlocal axis, we additionally probe an integral-type nonlocal regularization of the same in-house three-dimensional full-field reference (Ti-6Al-4V, 0°): the *local* damage model's own mesh dependence is large—refining from 27 to 64 grains moves its fracture strain by −15.92% (matching Supplementary Section S1) and shifts failure onto a localization branch—whereas the nonlocal operator compresses this to −1.8–4.0% (a factor of four to nine) at the fixed $D_{\mathrm{agg}} \geq D_c$ criterion, while raising ε_f monotonically with internal length (+2.0% at $\ell_c = 0.375h$ and +9.3% at one grain; internal-length scan and mesh-dependence table in Supplementary Section S1). This quantifies the strain-localization sensitivity that the local iso-strain surrogate cannot represent and confirms that the localization at which the spectral-FFT coupled iteration diverges is a boundary of the *local reference* rather than of the surrogate. A phase-field (Allen–Cahn) coupled reference, which would additionally require a redefinition of the fracture criterion and is not substituted by any proxy anywhere in this work, remains follow-on; its literature accuracy range is therefore quoted in Table 5 as qualitative context only, not as a like-for-like comparison. Reproducing the literature full-field approaches under the identical protocol would require full-field solvers that are identified as follow-on work in this environment (Section 7.3); the modified model is positioned explicitly as a low-order surrogate, and its blind-prediction statistics are not claimed to be competitive with full-field references.

Table 5: Comparison of fracture modeling approaches for AM alloys. Calibration and blind cross-validation errors are reported separately for the models computed in this work.

Model	**AM Features**	**Avg. ε_f Error**	**Computational Cost**
This work	4 features	20.4% (cal.) / 31.2% (blind)	Medium
Lemaitre CDM	None	18.6% (cal.) / 24.3% (blind)*	Low
CP + GTN (this work)	None	22.0% (cal.) / 36.7% (blind)*	Medium
CP + GTN [25]	Porosity only	∼15–20%	Medium
Phase-field [32]	None	∼10–15%	Very High
Macro CDM [47]	None	∼20–30%	Low

*All three rows are computed in this work under the identical leakage-free protocol (single fitted parameter per material, identical 20-grain Taylor aggregate, literature g_s): for the modified and Lemaitre models the fitted parameter is S_0, and for the CP+GTN baseline it is the constant void-nucleation rate A in $\dot{f} = A\dot{p}$ with fixed $f_c = 0.15$ (Taylor incompressibility makes the void-growth term vanish, so nucleation is the only void source; this strain-driven, defect-motivated form is chosen for mechanism contrast with the stress-driven Lemaitre form and is free of the threshold singularity of Gaussian nucleation). The GTN baseline carries no AM projection and predicts identical fracture strains at all three orientations, which is why its blind statistics (0/9 folds within 10%) support the value of the AM projection in the direction domain. "cal." is the calibration error, "blind" the leave-one-orientation-out blind prediction error. Literature comparisons (CP+GTN, phase-field, macro CDM) are indicative values from the cited works, not computed under the same protocol.

Table 5 situates the present work within the landscape of fracture modeling approaches. As a low-order surrogate, the proposed model offers a favourable

accuracy-to-cost trade-off by combining physically motivated AM corrections with an efficient scalar damage formulation. Among the three models computed under the identical protocol (Table 5), the AM projection is the only mechanism that carries the measured direction dependence (the GTN and Lemaitre baselines predict orientation-independent or weakened fracture strains), although its over-projection at 90° for the weakly anisotropic alloys degrades the blind statistics; the blind-prediction margin over the baseline is not statistically significant, so the framework is positioned as a physically motivated engineering tool with a strict, leakage-free validation protocol rather than as a high-precision prediction model.

7.8. Engineering implications

The calibrated and blind cross-validated model has direct implications for AM process and component design:

(1) **Process optimization**: By linking damage parameters to microstructural descriptors, the model enables virtual exploration of the process parameter space to identify conditions that minimize fracture susceptibility.
(2) **Build orientation selection**: The orientation-dependent f_{AM} provides quantitative guidance for selecting optimal build orientations to maximize fracture resistance.
(3) **Component qualification**: The standardized workflow reduces the experimental burden for component-specific fracture property determination, supporting AM part certification efforts.
(4) **Digital twin integration**: The model's explicit microstructure-property linkage makes it suitable for integration into digital twin frameworks for AM process monitoring and quality assurance.

8. Conclusions

This work presents a leakage-free validation framework, applied to a modified continuum damage mechanics model for fracture of additively manufactured metallic alloys, integrated within a crystal plasticity-based multiscale framework and implemented in the in-house Taylor-type crystal plasticity solver TaylorCPCDM, which follows DAMASK's constitutive interface conventions; DAMASK itself is used exclusively as an external full-field spectral-FFT reference: its damage module is deactivated ($S_0 \to \infty$) in every reported reference run, and the single attempt to activate a coupled damage model inside DAMASK diverged at damage onset and contributed no number, so all coupled-damage full-field references in this work come from the in-house two- and three-dimensional CPFEM rather than from DAMASK. The main contributions and findings are:

(1) **A strict, leakage-free validation methodology with a systematically mapped applicability domain**: The paper's primary

contribution is methodological. A four-tier verification framework separates code convergence, full-field homogenization-bias quantification, calibration/validation separation, and independent literature validation; the leakage-free leave-one-orientation-out protocol (no orientation multipliers, per-fold re-calibration of the single damage strength) is the only reported generalization statistic; the superseded orientation-multiplier figures (a near-interpolation calibration and a fixed-global-parameter sensitivity check) are documented in the Supplementary Material, Section S8 and are explicitly not blind predictions. Four independent blind sets (Inconel 718, and second datasets for Ti-6Al-4V, 316L and AlSi10Mg) are classified by failure mechanism into a decision procedure for model applicability (Section 7.2). Once every source angle is mapped to the build-direction convention, the *universal* blind failure is the non-transferability of S_0 across materials and process windows, *not* a direction reversal: a genuine 90° polarity reversal occurs in only one of the four sets (the defect-shielded Hitzler 316L), for which the geometric-shielding term $f_{\text{shield}}(\psi) = 1 - \beta_6 \sin^2 \psi$ repairs the 90° fold from 32.7% to 12.9% ($\beta_6 = 0.3$) without touching the calibration numbers ($\beta_6 = 0$; this β_6 scan is a post-hoc diagnostic excluded from every blind-prediction statistic); the second Ti-6Al-4V and AlSi10Mg sets fail in trend *shape* (a V-shaped datum against a monotonic law) and Inconel 718 in absolute *magnitude*, neither of which the shielding term addresses. The result is therefore a quantified boundary of validity. The extended-protocol statistics are reported at both of the dataset sizes they were computed at, and the one dataset that could be retrieved and integrated into them is reported as worsening the source-level statistic (51.98% to 64.13%) rather than as improving it. That boundary is further instantiated as a characterization-only, leakage-free gate ($\xi_0 \geq 3$) which routes each alloy either to the directional model or to the isotropic Lemaitre baseline and thereby recovers the per-alloy minimum of the two, giving a mean blind fracture-strain error of 15.9% against 24.3% for the always-isotropic baseline and 31.2% for the ungated directional model (Section 5.6, Table 3); the gate adds no fitted parameter, preserves the directional ratio (1.65 versus 1.67) wherever it invokes the descriptor, and is never worse than the baseline by construction. The gate is a fail-safe routing rule, not a validated operating curve, and it is delivered together with the protocol as the engineering screening rule of Section 7.2.

(2) **Directional information, not high precision**: Under the production leakage-free parameterization (single S_0 per material, literature g_s, directional correction $k_g = 0.15$, no orientation multipliers) the average *calibration* fracture-strain error is **20.4%** and the average UTS error **15.8%** (the corresponding near-interpolation calibration figures of the earlier parametrization are given in the Supplementary Material, Section S8). The strict leave-one-orientation-out blind error is **31.2%** (2/9 folds within the 10% target) versus **24.3%** for the Lemaitre baseline under the identical single-parameter protocol. The point estimate slightly favors the

baseline (the AM model differs by a mean +6.9 percentage points on the nine folds and +4.8 points on the expanded $n = 23$ set, i.e. nominally worse on both), and at equal parameter count the information criteria agree ($\Delta\text{AIC} = \Delta\text{BIC} = +10.8$ against the AM model at $k = 3$), so nothing in the central estimates supports the added descriptors as an accuracy improvement. The accuracy outcome is therefore reported strictly as a *failure to distinguish*, not demonstrated equivalence: on the $n = 23$ paired blind set the Bayes factor $\text{BF}_{01} \approx 1.8$–$5.8$ (prior-width dependent) gives at most weak-to-moderate support for the absence of a mean difference, and a two-one-sided equivalence test rejects non-equivalence at no margin from ± 10 to ± 25 points (Section 5, Supplementary Section S11). The added value of the AM projection is therefore purely directional: it matches the strongly columnar Ti-6Al-4V direction ratio quantitatively (1.65 versus 1.67) and carries the measured direction ordering that direction-agnostic baselines lack—a mean Kendall $\bar{\tau} = +0.21$ with a bootstrap 95% CI of $[-0.33, +0.71]$ over eight source-material groups, i.e a positive but not individually significant point estimate (the equal-parameter reference is itself mildly direction-*incorrect* on the same groups, $\bar{\tau}_{\text{lem}} = -0.21$, so the descriptor's ordering value is measured against a direction-wrong rather than a direction-neutral baseline) whose near-quantitative strength is localized to the correctly-polarised columnar domain that the gate of contribution (1) isolates (hierarchical directional-gain posterior, Section 6); its over-projection of the weakly anisotropic FCC alloys (316L ratio 2.42 versus 1.45; AlSi10Mg 2.40 versus 1.51) is the leading quantified residual, further traced to a projection-normalization artifact and evaluated as a targeted relative-normalization correction that removes the 316L sign reversal at the factor level but does not improve the blind average or the model-selection outcome (Section 5.5).

(3) **A physically-motivated CDM modification**: The damage evolution equation explicitly incorporates five AM-specific microstructural descriptors—porosity volume fraction (ϕ), initial damage (D_0), melt pool boundary density (λ), grain morphology factor (ξ), and texture orientation weight (θ)—through the multiplicative AM correction function f_{AM}, each component with a clear physical interpretation grounded in the mechanisms of damage nucleation and propagation in AM alloys.

(4) **One-way damage-degraded CP-CDM coupling**: The modified damage model is implemented at the constitutive level in our in-house Taylor-type solver TaylorCPCDM, which follows DAMASK's constitutive interface conventions; the one-way coupling enables concurrent evolution of crystallographic slip and damage at each integration point, with damage degrading the nominal (measured) stress while the slip-driving stress is evaluated on the undamaged stiffness. A full two-way feedback is deliberately not claimed; its absence is quantified in Section 7 as a limitation alongside the Taylor-homogenization bias. DAMASK itself is used exclusively as an external full-field spectral-FFT reference for the damage-free bias, and the coupled-damage fracture-strain bias is quantified against

in-house two-dimensional plane-strain and $3 \times 3 \times 3$ three-dimensional CPFEM references under identical settings (Section 7.3); the extension of the three-dimensional reference to an integral-type nonlocal damage regularization, and its effect on the mesh dependence and the 0° agreement with the Taylor surrogate, are reported in full in Supplementary Section S1.

(5) **Quantified Taylor-homogenization bias, resolved through the plastic-flow branch**: A full-field DAMASK spectral-FFT reference on the identical 30-grain microstructure shows that the Taylor iso-strain assumption overestimates the damage-free UTS by 16.5–16.7% for the FCC alloys and underestimates it by 12.0% for Ti-6Al-4V (Section 7.3). The bias is resolved through the whole plastic-flow branch rather than only at the UTS and fracture strain: the flow-stress and hardening-rate biases are quantified per material in Table S30, and they grow with plastic strain for the FCC alloys, so the iso-strain rule distorts the *shape* of the flow curve and not merely its level (Fig. S8). The Taylor-based predictions are therefore positioned as a low-order surrogate model, and the bias range is reported as an explicit uncertainty source. The band is quantified for four build orientations in Section 7.3 and revisited under a nonlocal regularization of the same reference in Supplementary Section S1, where the sign-bracketing structure of the band is shown to be a property of the local reference at the present internal length rather than a robust feature of the surrogate.

(6) **Open-source standardized workflow**: An end-to-end automated simulation pipeline is established and released as open-source software, facilitating adoption and reproducibility.

Scope and external-validation caveat. No original experimental campaign was conducted in this work: all calibration and validation data are literature-reported measurements compiled for comparable LPBF process windows (Section 5.2); the independent out-of-source assessment rests on literature blind tests from separate publications (Inconel 718, a second Ti-6Al-4V dataset, a second 316L dataset, and a second AlSi10Mg dataset, Sections 7.4–7.5), which are the only out-of-source checks available. The reported calibration errors are internal to the primary literature dataset, and extrapolation to untested materials, process parameters, or loading modes carries the explicit risk quantified by the blind LOOCV error (31.2%), the Taylor-homogenization bias (16.5–16.7% overestimation for FCC, 12.0% underestimation for Ti-6Al-4V in the damage-free stage) and the coupled-damage fracture-strain bias (-26.5 to -35.8%, Ti-6Al-4V; +5.3%, AlSi10Mg) quantified against the in-house CPFEM reference (Section 7.3); the model is accordingly positioned as a physically motivated, low-order surrogate with quantified limitations, and original experiments and phase-field coupled validation are identified as the priority for follow-on work; an independent external finite-element engine (`scikit-fem` 12) has been used to cross-check the in-house two-dimensional coupled-damage reference (Supplementary Section S13), confirming the framework-independence

of the Ti-6Al-4V localisation trend while the AlSi10Mg localisation criterion remains mesh- and solver-sensitive—a discretisation cross-check, not a phase-field calculation; a first three-dimensional CPFEM coupled-damage reference has been completed in this environment (Section 7.3) and now carries an integral-type nonlocal regularization whose measured effect on that reference's mesh dependence is reported in Supplementary Section S1; the effect is a compression of the mesh dependence and not its removal, and it does not narrow the surrogate's disagreement with the reference.

Future work will extend the model in several directions: (i) incorporating a tensorial damage formulation to capture directional stiffness degradation; (ii) establishing explicit process-structure-property linkages for direct process parameter optimization; (iii) developing reduced-order surrogate models for computationally efficient industrial-scale simulations; and (iv) extending validation to additional loading modes (compression, shear, cyclic) and AM process conditions.

Acknowledgments

Not applicable.

Declaration of competing interest

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

This work was supported by the "Pioneers and Leaders + X" R&D Plan of the Zhejiang Provincial Major Sci-Tech Innovation Platform Project [grant number 2025SSYS0006], corresponding to the Key Research Project of Zhejiang Lab [grant number 3700-3AA250100].

Data Availability

The simulation workflow, including input files, calibration scripts, and post-processing tools, is released as open source and available at the project repository (https://github.com/WangHan2304/am-fracture-cpcdm), archived under a versioned DOI at https://doi.org/10.5281/zenodo.23257091. The calibrated material parameters are provided in Table 1; the raw outputs of all simulations (stress–strain curves, damage trajectories, and calibration tables) are archived in the repository. The in-house Taylor-type solver (TaylorCPCDM), the two- and three-dimensional coupled-damage full-field CPFEM reference scripts, and the leakage-free cross-validation cache and its expanded paired-blind, Sobol-sensitivity, model-simplification and statistical-upgrade drivers are all released with the code, together

with `REPRODUCIBLE_FULLFIELD_REFERENCE.md` documenting the fixed seed, mesh and increment needed to regenerate every full-field and blind number. The third-party finite-element cross-check, the directional-gain hierarchical-Bayes posterior (Section 6, Supplementary Section S7) and the Bayes-factor/TOST/bootstrap-Kendall statistical upgrade (Section 5.5, Supplementary Section S11) are each released with their driver scripts and result JSONs, and the external `scikit-fem` engine together with the constrained install environment (the conda/`fenics-dolfinx` routes being blocked) is documented in Supplementary Sections S13 and S9; all three are purely post-hoc re-analyses that re-use stored predictions (64 walkers, seed 42; $n = 23$ paired blind set) and introduce no new fitted parameter or production number. All simulation data supporting the findings of this study are available from the corresponding author upon reasonable request.

Reproducibility manifest.. Content-hash anchoring of every reported number to the single script and result-JSON field that recompute it, the locked environment (Python 3.13.0, NumPy 2.3.4, SciPy 1.17.1, seed 42) and the full number-to-source binding table are documented in the Supplementary Material, Sections S3 and S4, and archived with the released repository.

All experimental data used in this work are literature-reported measurements, not original experiments: the nine primary cases are compiled representative values from the cited publications, and the extended dataset of Section 5.2 is a curated compilation of further literature values whose per-point provenance (source, figure or table, and the orientation conversion applied) is recorded in the extraction log archived with the code. For the 30 points taken from the NIST mds2-3402 release, which is public under the NIST Open License, nothing is redistributed: only the source's own published group means are tabulated, the archive's checksum verification and the orientation mapping are recorded in the extraction log, and every value remains traceable to a numbered table of the accompanying report. No public benchmark measured data is redistributed or re-reported here; the assessment of the public AM dataset hosts, including the finding that NIST AM-Bench contains no Ti-6Al-4V or AlSi10Mg tensile data, the inaccessibility of America Makes and NIMS MatNavi, and the resolution of the NIST repository access limitation by retrieving mds2-3402 through its object-store endpoint, is documented in Section 7.6.

CRediT authorship contribution statement

Han Wang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft. **Lu Zhang:** Methodology, Validation, Writing – review & editing, Supervision. **Anna Stepashkina:** Resources, Validation, Writing – review & editing. **Qi Hu:** Supervision, Project administration, Writing – review & editing. All authors read and approved the final manuscript.

References

[1] T. DebRoy, H.L. Wei, J.S. Zuback, T. Mukherjee, J.W. Elmer, J.O. Milewski, A.M. Beese, A. Wilson-Heid, A. De, W. Zhang, "Additive manufacturing of metallic components – Process, structure and properties," *Progress in Materials Science*, vol. 92, pp. 112–224, 2018.

[2] D. Herzog, V. Seyda, E. Wycisk, C. Emmelmann, "Additive manufacturing of metals," *Acta Materialia*, vol. 117, pp. 371–392, 2016.

[3] M. Seifi, A. Salem, J. Beuth, O. Harrysson, J.J. Lewandowski, "Overview of materials qualification needs for metal additive manufacturing," *JOM*, vol. 68, pp. 747–764, 2016.

[4] L. Thijs, F. Verhaeghe, T. Craeghs, J. Van Humbeeck, J.-P. Kruth, "A study of the microstructural evolution during selective laser melting of Ti–6Al–4V," *Acta Materialia*, vol. 58, pp. 3303–3312, 2010.

[5] J. Suryawanshi, K.G. Prashanth, U. Ramamurty, "Mechanical behavior of selective laser melted 316L stainless steel," *Materials Science and Engineering: A*, vol. 696, pp. 113–121, 2017.

[6] B.E. Carroll, T.A. Palmer, A.M. Beese, "Anisotropic tensile behavior of Ti–6Al–4V components fabricated with directed energy deposition additive manufacturing," *Acta Materialia*, vol. 87, pp. 309–320, 2015.

[7] Y. Kok, X.P. Tan, P. Wang, M.L.S. Nai, N.H. Loh, E. Liu, S.B. Tor, "Anisotropy and heterogeneity of microstructure and mechanical properties in metal additive manufacturing: A critical review," *Materials & Design*, vol. 139, pp. 565–586, 2018.

[8] M. Simonelli, Y.Y. Tse, C. Tuck, "On the texture formation of selective laser melted Ti–6Al–4V," *Metallurgical and Materials Transactions A*, vol. 45, pp. 2863–2872, 2014.

[9] S. Leuders, M. Thöne, A. Riemer, T. Niendorf, T. Tröster, H.A. Richard, H.J. Maier, "On the mechanical behaviour of titanium alloy TiAl6V4 manufactured by selective laser melting: Fatigue resistance and crack growth performance," *International Journal of Fatigue*, vol. 48, pp. 300–307, 2013.

[10] W. Wen, Y. Liu, Y. Chen, J. Ma, "Damage evolution and fracture of additively manufactured 316L stainless steel under monotonic tension," *Materials Science and Engineering: A*, vol. 764, p. 138232, 2019.

[11] G.R. Johnson, W.H. Cook, "Fracture characteristics of three metals subjected to various strains, strain rates, temperatures and pressures," *Engineering Fracture Mechanics*, vol. 21, pp. 31–48, 1985.

[12] Y. Bai, T. Wierzbicki, "A new model of metal plasticity and fracture with pressure and Lode dependence," *International Journal of Plasticity*, vol. 24, pp. 1071–1096, 2008.

[13] X. Shang, Z. Cui, M.W. Fu, "A ductile fracture model considering stress state and Zener-Hollomon parameter for hot deformation of metallic materials," *International Journal of Mechanical Sciences*, vol. 144, pp. 800–812, 2018.

[14] A.L. Gurson, "Continuum theory of ductile rupture by void nucleation and growth: Part I – Yield criteria and flow rules for porous ductile media," *Journal of Engineering Materials and Technology*, vol. 99, pp. 2–15, 1977.

[15] V. Tvergaard, A. Needleman, "Analysis of the cup-cone fracture in a round tensile bar," *Acta Metallurgica*, vol. 32, pp. 157–169, 1984.

[16] Z. Zhang, T.-S. Jun, T.B. Britton, F.P.E. Dunne, "Determination of Ti-6242 α and β slip properties using micro-pillar test and computational crystal plasticity," *Journal of the Mechanics and Physics of Solids*, vol. 95, pp. 393–410, 2016.

[17] J. Liu, J. Li, Y. Liu, Z. Moumni, "Crystal plasticity modeling of the mechanical behavior of additively manufactured 316L stainless steel," *International Journal of Plasticity*, vol. 133, p. 102793, 2020.

[18] K. Kapoor, R. Sangid, A.M. Beese, "Modeling the anisotropic mechanical behavior of additively manufactured Ti–6Al–4V using crystal plasticity finite element methods," *Additive Manufacturing*, vol. 47, p. 102334, 2021.

[19] J. Lemaitre, *A Course on Damage Mechanics*, Springer-Verlag, Berlin, 1992.

[20] J.L. Chaboche, "Continuum damage mechanics: Part I – General concepts; Part II – Damage growth, crack initiation, and crack growth," *Journal of Applied Mechanics*, vol. 55, pp. 59–72, 1988.

[21] J. Lemaitre, "A continuous damage mechanics model for ductile fracture," *Journal of Engineering Materials and Technology*, vol. 107, pp. 83–89, 1985.

[22] J. Dufailly, J. Lemaitre, "Modeling very low cycle fatigue," *International Journal of Damage Mechanics*, vol. 4, no. 2, pp. 153–170, 1995.

[23] S. Kasherman, M. Brandt, M. Easton, "Porosity in additively manufactured metals: A review," *Metals*, vol. 10, p. 1296, 2020.

[24] A. Needleman, "An analysis of ductile rupture modes at a crack tip," *Journal of the Mechanics and Physics of Solids*, vol. 35, pp. 151–183, 1987.

[25] Y. Zhang, J. Chen, L. Liu, "Modeling of anisotropic ductile fracture in additively manufactured Ti-6Al-4V using crystal plasticity coupled GTN model," *International Journal of Plasticity*, vol. 143, p. 103018, 2021.

[26] J.D. Eshelby, "The determination of the elastic field of an ellipsoidal inclusion, and related problems," *Proceedings of the Royal Society of London A*, vol. 241, pp. 376–396, 1957.

[27] J. Köster, S. Rahmann, "Snakemake—a scalable bioinformatics workflow engine," *Bioinformatics*, vol. 28, pp. 2520–2522, 2012.

[28] F. Roters, M. Diehl, P. Shanthraj, P. Eisenlohr, C. Reuber, S.L. Wong, T. Maiti, A. Ebrahimi, T. Hochrainer, H.-O. Fabritius, S. Nikolov, M. Friák, N. Fujita, N. Grilli, K.G.F. Janssens, N. Jia, P.J.J. Kok, D. Ma, F. Meier, E. Werner, M. Stricker, D. Weygand, D. Raabe, "DAMASK – The Düsseldorf Advanced Material Simulation Kit for modeling multi-physics crystal plasticity, thermal, and damage phenomena from the single crystal up to the component scale," *Computational Materials Science*, vol. 158, pp. 420–478, 2019.

[29] Z. Wang, M. Xu, X. Liu, Q. Lin, X. Huang, H. Zhang, "Experimental and crystal plasticity finite element investigations of plastic anisotropy in additively manufactured Ti6Al4V alloy," *Metals*, vol. 14, no. 1, p. 130, 2024. doi:10.3390/met14010130.

[30] M. Jaskari, et al., "Tensile properties and deformation of AISI 316L additively manufactured with various energy densities," *Materials*, vol. 14, no. 19, p. 5809, 2021. doi:10.3390/ma14195809.

[31] Z. Wu, S. Wu, X. Gao, et al., "The role of internal defects on anisotropic tensile failure of L-PBF AlSi10Mg alloys," *Scientific Reports*, vol. 13, p. 14681, 2023. doi:10.1038/s41598-023-39948-z.

[32] C. Liu, A. Rehman, M. Bambach, "Phase-field fracture modeling of anisotropic ductile damage in additively manufactured 316L stainless steel," *Computational Materials Science*, vol. 213, p. 111639, 2022.

[33] T. Sun, et al., "Anisotropic mechanical properties of laser powder bed fused Ti-6Al-4V," *Advanced Engineering Materials*, vol. 26, 2024. doi:10.1002/adem.202400942.

[34] L. Hitzler, J. Hirsch, B. Heine, M. Merkel, W. Hall, A. Oechsner, "On the anisotropic mechanical properties of selective laser-melted stainless steel," *Materials*, vol. 10, no. 10, p. 1136, 2017. doi:10.3390/ma10101136.

[35] M. Awd, F. Stern, A. Kampmann, D. Kotzem, J. Tenkamp, F. Walther, Microstructural characterization of the anisotropy and cyclic deformation behavior of selective laser melted AlSi10Mg structures, Metals 8 (10) (2018) 825.

[36] E. W. Hovig and A. S. Azar, "Determination of anisotropic mechanical properties for materials processed by laser powder bed fusion," *Advances in Materials Science and Engineering*, vol. 2018, p. 7650303, 2018. doi:10.1155/2018/7650303.

[37] D.S. Watring, J.T. Benzing, N. Hrabe, A.D. Spear, "Effects of laser-energy density and build orientation on the structure–property relationships in

as-built Inconel 718 manufactured by laser powder bed fusion," *Additive Manufacturing*, vol. 36, p. 101425, 2020. doi:10.1016/j.addma.2020.101425.

[38] N. Derimow, J.T. Benzing, N. Moser, O.L. Kafka, H. Joress, A. McDannald, D. Newton, L. Koepke, N. Hrabe, "Dataset of additive manufacturing laser powder bed fusion Ti-6Al-4V subjected to various novel hot isostatic pressing (HIP) treatments," NIST Public Data Repository, ARK `ark:/88434/mds2-3402`, version 1.0.0 (2024-07-09) and 1.0.2 (2026-01-06), NIST Open License, public. doi:10.18434/mds2-3402.

[39] N. Derimow, J.T. Benzing, N. Moser, O.L. Kafka, H. Joress, A. McDannald, D. Newton, L. Koepke, N. Hrabe, "High pressure heat treatment development for laser powder bed fusion Ti-6Al-4V alloy: observations and findings," NIST Internal Report (IR) 8551, December 2024. doi:10.6028/NIST.IR.8551.

[40] A. Venkatraman, C.E. Johnson, D.L. McDowell, S.R. Kalidindi, "Bayesian protocols for high-throughput identification of kinematic hardening model forms," *International Journal of Mechanical Sciences*, vol. 285, p. 109791, 2025.

[41] S. Murakami, "Mechanical modeling of material damage," *Journal of Applied Mechanics*, vol. 55, pp. 280–286, 1988.

[42] P. Eisenlohr, M. Diehl, R.A. Lebensohn, F. Roters, "A spectral method solution to crystal elasto-viscoplasticity at finite strains," *International Journal of Plasticity*, vol. 46, pp. 37–53, 2013.

[43] P. Shanthraj, B. Svendsen, L. Sharma, F. Roters, D. Raabe, "Elasto-viscoplastic phase field modelling of anisotropic cleavage fracture," *Journal of the Mechanics and Physics of Solids*, vol. 99, pp. 19–34, 2017.

[44] "Impact of heat treatment and building direction on selective laser-melted 316L stainless steel: microstructure and mechanical properties," PubMed Central record PMC11721696, 2024. https://pmc.ncbi.nlm.nih.gov/articles/PMC11721696.

[45] F. M. Andrade Pires, J. M. A. César de Sá, "Assessment and comparison of non-local integral models for ductile damage," *International Journal of Damage Mechanics*, 2014. doi:10.1177/1056789513493103.

[46] S. Satouri, G. Chatzigeorgiou, A. Benaarbia, F. Meraghni, "A gradient-enhanced constitutive framework for the investigation of ductile damage localization within semicrystalline polymers," *International Journal of Damage Mechanics*, vol. 31, no. 10, 2022.

[47] X. Chen, Y. Li, W. Zhang, "Macroscopic continuum damage mechanics modeling of ductile fracture in additively manufactured metals," *Engineering Fracture Mechanics*, vol. 260, p. 108195, 2022.